

The asymptotic analysis of a junction of two elastic beams

G. Leugering¹ | S. A. Nazarov^{2,3} | A. S. Slutskiy^{2,3}

¹Department Mathematik Friedrich-Alexander-Universität Erlangen-Nürnberg, Cauerstr. 11 (03.322), 91058 Erlangen

²Saint Petersburg State University, St. Petersburg State University, 7/9 Universitetskaya nab., St. Petersburg, 199034, Russia

³Institute of Problems of Mechanical Engineering V.O., Bolshoj pr., 61 St. Petersburg, 199178, Russia

Correspondence

G. Leugering, Department Mathematik Friedrich-Alexander-Universität Erlangen-Nürnberg, Cauerstr. 11 (03.322), 91058 Erlangen.

Email: guenter.leugering@fau.de

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This paper is devoted to describing the asymptotic behavior of a structure consisting of thin elastic planar beams coupled at flexible joints. As the thickness of the beams tends to zero, we establish classical 1-D beam equations for each individual structural element and transmission conditions across the joints.

KEYWORDS

asymptotic analysis, plane systems of beams, thin elastic junctions

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1 | INTRODUCTION

1.1 | Motivation

Flexible beam structures are crucial in many applications. In car and aerospace industry, the design of appropriate light weight frame structures is a key for modern product development. Flexible beam structures also appear on a macroscopic level in civil engineering, where, e.g. frames for buildings and bridges play a crucial role. On the microscopic level, the understanding of beam networks is important for the design of smart materials. While the dynamical behaviour of the structural elements, like beams, shells, plates or cables, are very well known since a long time, the interaction of these elements across flexible joints is still of great interest. The literature on beam structures is overwhelming and we do not even try to account for it. There are basically two approaches for the modeling of such structures. The first approach is based on a direct modeling in the sense that variational methods are used to derive the corresponding transmission conditions for a particular 1-D beam model. In particular, Euler-Bernoulli or Timoshenko beam equations are taken for each individual structural element of the network together with continuity and geometric conditions across the nodes. Then Krichhoff-type force and moment conditions are established via variational calculus. See the article by Horne^[4] for an early exposition of the variational transmission conditions and Bathe^[1] for an early finite element realization of frame structures. Mathematically rigorous analysis dates back to the late 1980ies; see e.g. [3]. A general rigorous network approach for string and beam structures has been pursued in [5].

The second approach is based on asymptotic analysis. It starts from a network-like elastic 2-D or 3-D body and aims at a thin-domain approximation. Examples of this approach are provided by the collection of articles in [2] and, in particular, by [6] for a cantilever requiring for certain symmetries and [14,15] for general configurations.

In [6], the author considered isotropic elastic material in a rectangular 3-D-configuration (a carpenter's square) with a corresponding 1-D-skeleton with one end clamped and the other end free. The method consists of scaling the Lamé moduli and

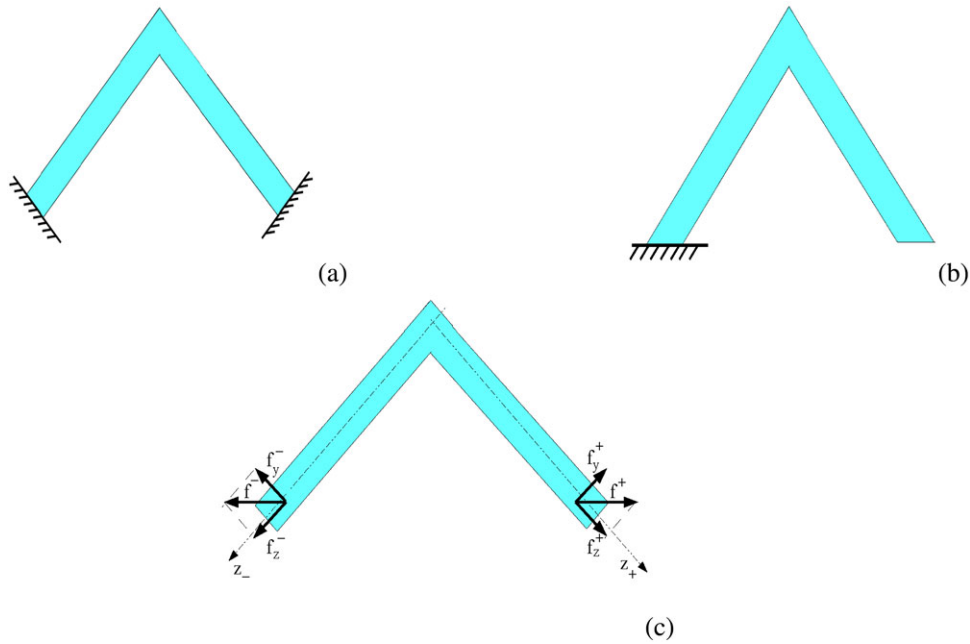


FIGURE 1 Junction $\Omega(h)$ with (a) two clamped legs, (b) one clamped leg and (c) free legs

establishing an H^1 -norm bound (a version of Korn's inequality), independent of the small thickness of the structure. The corresponding asymptotic analysis leads to infinite longitudinal stiffness of the beams involved, an effect that significantly influences the transmission conditions for the resulting 1-D cantilever. In the asymptotic limit, the author obtains two Euler-Bernoulli beams with transverse and lateral displacements and coupling conditions at the connecting node that are similar to those obtained in [5] if one lets the longitudinal stiffness grow to infinity.

In [14,15] so-called *slowly movable* and *movable parts* have been introduced which appear e.g. in a portal frame structure. In these publications, the material may be anisotropic elastic and the beam geometries are general. Again, a generalized Korn's inequality for the overall structure, independent of the thickness of the original thin body, plays a major role. The asymptotic ansatz is made with for the solutions with respect to the thickness h in the orders starting from h^{-2} . Then, for each order, successively boundary value problems have to be solved such that, finally, in the asymptotic limit one obtains a coupled second-order and fourth-order system that represent the longitudinal extension and bending of the beams elements as well as corresponding boundary and nodal conditions, reflecting the slowly movable and movable parts of the structure. Those conditions involve algebraic unknowns which describe longitudinal displacement of movable elements. In [14], the geometrical nodal conditions have been established. In the current article, the authors also rigorously develop the transmission conditions of Kirchhoff-type and the non-local nature of the conditions for movable parts that appears after excluding the algebraic unknowns from the limiting one-dimensional problem.

1.2 | Description of the asymptotic ansatz and outline of the paper

We first consider a planar, linear elastic, non-isotropic body $\Omega(h)$ consisting of two 'thick' beams $Q(h) = \{x = (x_1, x_2) : |x_1| < 1/2, |x_2| < h/2\}$ and a junction region; see Figure 1 for a sketch and Figure 2, (1.16), (1.17) for the precise definition. On $\Omega(h)$, let the displacement vector $u(h, x) = u^h(x)$ solve the boundary value problem (1.21), (1.22). We ask for a limiting problem and the corresponding solution, as the thickness parameter tends to zero. We are particularly interested in the coupling conditions at the emerging *thin junction*. To this end, in Section 2, we propose an asymptotic ansatz (i.e. (2.1)) for u^h and invoke in Section 3 a variant of Korn's inequality (3.6) which is independent of h . This will enable us to prove error estimates for the discrepancies between the solution u^h of the original boundary value problem and the solution of the limiting system. Once this problem is settled, in Section 4, we analyze the situation for more general beam structures.

In order to be more specific, in the Kirchhoff theory of thin elastic plates and beams, the averaged displacement vector w^h , depending on the relative thickness $h \ll 1$ of the beam $Q(h)$ and normalized in an appropriate way, admits different orders, say, h^{-2} and h^{-1} , of magnitudes in the transverse e_2 and longitudinal e_1 directions, respectively. This remains valid for rather arbitrary loadings, however, for a single beam with either clamped, or free ends. A general classification^[14,15] of thin elastic

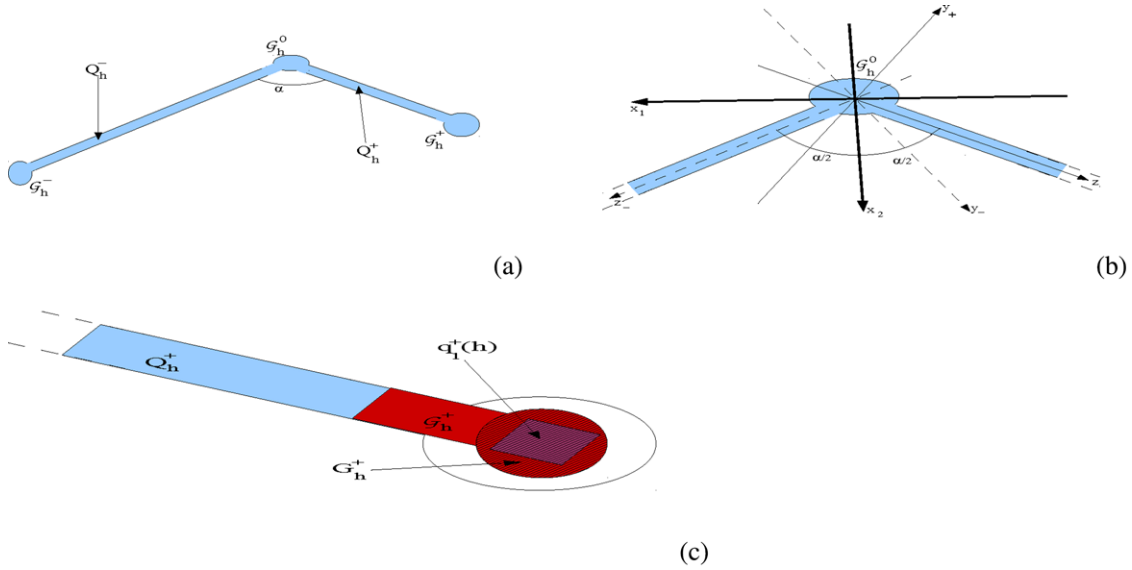


FIGURE 2 (a) The junction $\Omega(h)$ and (b) the coordinates (x_1, x_2) and $(y_{\pm}, z_{\pm})$. (c) The augmented knot G_h^+

junctions predicts so-called *movable* parts, for which both components of the displacement vector exhibit the same order h^{-2} such that the traditional main asymptotic term of the displacement vector u^h changes from

$$\begin{aligned} u^h(x) &\sim h^{-2} e_2 w_2(x_1) \quad \text{to} \\ u^h(x) &\sim h^{-2} e_2 w_2(x_1) + h^{-2} e_1 a_1, \end{aligned} \quad (1.1)$$

where w_2 is a function in $x_1 \in (-1/2, 1/2)$ and a_1 is a constant (see (2.1) and (2.3),(2.2) for the complete ansatz). The additional constant term on the right hand side of (1.1) vanishes upon differentiation and, therefore, does not directly affect the asymptotics of strains and stresses in $Q(h)$. At the same time, the nodal conditions of the limiting one-dimensional problem for an elastic junction with movable parts involves algebraic components of the type a_1 in (1.1) and, thus, the whole strain-stress description of the junction becomes crucially dependent on the algebraic components of the asymptotic ansatz for movable beams.

The general classification procedure mentioned above is recursive and in a way tedious so that the first goal of our paper is to demonstrate a simplified version for an elementary junction $\Omega(h)$ of elastic beams in the shape of the letter Λ without any assumption on physical and geometrical symmetries. In this introduction, we assume that the “legs” $Q^{\pm}(h)$ are rectangles of size $l_{\pm} \times h$, which are coupled at a right angle, although in the main body of the article, we consider an arbitrary angle between the beams that allows e.g. for shape optimization. Deviating further from the pioneering result,^[6] we also deal with anisotropic and inhomogeneous materials including laminate ones as the limiting case (cf. [12]). However, the main goal of our study is to detect particular cases of beam junctions and exterior loadings which allow for asymptotic ‘ansätze’ leading to the one-dimensional models established in [5].

If both legs of the junction are fixed at their roots (external ends) as in Figure 1(a), both beams $Q^{\pm}(h)$ are *hard-movable* according to [14] and the averaged displacement vectors $w^{\pm}$ in the ansatz

$$u^h(x) \sim h^{-2} e_2^{\pm} w_2^{\pm}(z_{\pm}) + h^{-1} e_1^{\pm} \left(w_1^{\pm}(z_{\pm}) - y^{\pm} \partial_{z_{\pm}} w_2^{\pm}(z_{\pm}) \right) + \dots, \quad x \in Q^{\pm}(h), \quad (1.2)$$

(see the notation in Figure 1(c) and Figure 2(b), satisfy the boundary conditions at the external end points $z_{\pm} = l_{\pm}$

$$w_2^{\pm}(l_{\pm}) = 0, \quad w_1^{\pm}(l_{\pm}) = 0, \quad \partial_{z_{\pm}} w_2^{\pm}(l_{\pm}) = 0 \quad (1.3)$$

as well as the *stable* transmission conditions (for the terminology see [8]) at the junction point

$$w_2^{\pm}(0) = 0, \quad \partial_{z_+} w_2^+(0) = \partial_{z_-} w_2^-(0). \quad (1.4)$$

Conditions (1.3) and (1.4) are built into the function space in the variational formulation of the limiting problem, while the other three *intrinsic* transmission conditions can be derived with the help of the Green formula, see [8] and cf. Section 2.4, and imply the continuity of the torque $T^\pm$ and the annihilation of the longitudinal forces $F_1^\pm$ at the interior node, namely

$$T^+(0) + T^-(0) = 0, \quad F_1^\pm(0) = 0. \quad (1.5)$$

This limiting one-dimensional problem is uniquely solvable and its solution $w^\pm$ provides the main terms in the asymptotic expansion (1.2) and, after proper differentiation, in the asymptotics of the stresses. The asymptotic remainders can be estimated in the energy norm by means of the anisotropic weighted inequality of Korn's type, cf. [9,14]. These conclusion have been originally made in the advanced paper,^[6] see also [7], but also follow from general results in [14,15], due to the obvious classification of elements in the junction $\Omega(h)$.

For beams with free exterior ends (Figure 1(b) and (c)), the ansatz and the transmission conditions encounter crucial changes. In the case of the junction in Figure 1(b) the ansatz (1.2) is kept at the clamped beam $Q^-(h)$ while the next beam $Q^+(h)$ is attached to the free end of the first beam $Q^-(h)$ that enjoys a transversal shift of order h^{-2} which becomes the longitudinal shift for the beam $Q^+(h)$ without clamped ends. This is just the reason for the change (1.1) in the ansätze and introducing a constant a_1^+ in the longitudinal displacement. This constant influences both, the transmission condition and the boundary conditions at the free end of the beam $Q^+(h)$.

In the case of junction with both free legs (Figure 1(c)), in the ansatz (1.2) augmented with the terms $h^{-2}e_1^\pm a_1^\pm$ according to (1.1), we have to impose the relations

$$w_2^\pm(0) = a_1^\mp, \quad (1.6)$$

which are caused by the continuity of the translations at the junction point $z_\pm = 0$ and imply nothing but the determination of the constants $a_1^\pm$ through the bending part $w_2^\mp$. In this way, we are left with only one stable transmission condition from the list (1.4), that is, the continuity of the rotation

$$\partial_z w_2^+(0) = \partial_z w_2^-(0). \quad (1.7)$$

Hence, all other five transmission conditions at $z_\pm = 0$, as well as three boundary conditions at $z_\pm = l$, are intrinsic (in the terminology of [8]) and, therefore, disappear from the variational formulation. Since the latter contains a formally positive elliptic system for the components $w_1^\pm$ and $w_2^\pm$ of the compound vector

$$w = (w^+, w^-)^\top = (w_1^+, w_2^+, w_1^-, w_2^-)^\top, \quad (1.8)$$

we, at the first sight, reveal five linear independent solutions of the homogeneous limiting problem:

$$w^{(1)} = (0, 1, 0, 0)^\top, \quad w^{(2)} = (0, 0, 0, 1)^\top, \quad w^{(3)} = (0, z_+, 0, z_-)^\top, \quad (1.9)$$

$$w^{(4)} = (1, 0, 0, 0)^\top, \quad w^{(5)} = (0, 0, 1, 0)^\top. \quad (1.10)$$

However, according to [14,15], the presence of arbitrary constants $a_1^\pm e_1^\pm$ in the main term of order h^{-2} entails the orthogonality conditions

$$\int_0^{l_\pm} w_1^\pm(z) dz = 0 \quad (1.11)$$

for the higher-order terms. Thus, the last couple (1.10) has to be removed from the list and we obtain just three solutions as in the original two-dimensional elasticity problem in $\Omega(h)$, that is, rigid motions. Indeed, one easily finds out in (1.9) the prototypes for two translations and one rotation. Consequently, we have three compatibility conditions in the limiting problem. By similar reasoning as in, e.g., [10, Ch.4 and 16], these conditions must be inherited from the compatibility conditions (note that these are also three!) in the original problem in $\Omega(h)$. As another situation, we assume that, as in Section 2.6, the junction is stretched by the forces $f^\pm$ depicted in Figure 1(c), and applied at the bases of the legs $Q^\pm(h)$. Owing to the Saint-Venant principle, the boundary conditions in the limit problem read:

$$F_2^\pm(l_\pm) = f_y^\pm, \quad F_1^\pm(l_\pm) = 0, \quad T^\pm(l_\pm) = 0, \quad (1.12)$$

where $f_y^\pm$ is the projection of $f^\pm$ on the $y^\pm$ -axis. Since it is much easier to bend a beam than to stretch it, the other projections $f_{z_\pm}^\pm$, being of the same order in h as $f_y^\pm$, cannot enter the limiting problem directly, but are needed for the self-balance of external actions. A permeation mechanism for these projections looks as follows: in addition to the terms $h^{-2}a_1^\pm e_1^\pm$, caused by the primary substitution (1.1), we extend the asymptotic ansatz (1.2) with the new higher-order terms

$$h^0 W^\pm(z_\pm) e_1^\pm, \quad (1.13)$$

such that the longitudinal forces generated by (1.13) equal $f_z^\pm$ and simultaneously are transported from the exterior end $z_\pm = l$ to the interior end $z_\pm = 0$. Finally, the *longitudinal* force along $Q^\pm(h)$ is balanced by a *transversal* force applied to the end $z^\mp = 0$ of the other beam $Q^\mp(h)$ and in this way $f_z^\pm$ appears in the intrinsic transmission conditions

$$F_2^\pm(0) = f_z^\pm, \quad F_1^\pm(0) = 0, \quad T^+(0) + T^-(0) = 0. \quad (1.14)$$

The nodal conditions (1.7), (1.14) and (1.12) close the systems of ordinary differential equations for the vector function (1.8), while the necessary compatibility conditions are fulfilled due to the self-balance of the exterior loading on $\Omega(h)$. Formulas (1.6) and (1.11) specify entries in the advanced asymptotic ansatz. It is remarkable that the introduced asymptotic terms (1.13) are beyond the asymptotic precision of the derived limit problem and, in this way, within the Kirchhoff model, the discussed projection $f_z^\pm$ seems to jump over spontaneously from the exterior end of the beam $Q^\pm(h)$ to the interior end of the other beam $Q^\mp(h)$. It should be mentioned that within the approach^[5] this effect of “far-interaction” does not appear.

1.3 | The description of the elastic junction

In the plane $\mathbf{R}^2$, we consider the junction $\Omega(h)$ of two thin beams with angle $\alpha \in (0, \pi)$ between them, see Figure 2(a). In order to describe the shape of this junction, we introduce the Cartesian coordinates (x_1, x_2) and $(y_\pm, z_\pm)$ with the common origin O (see Figure 2(b)). The axes $z_\pm$ directed along the beams

$$Q^\pm(h) = \{(y_\pm, z_\pm) \in \mathbf{R}^2 : z \in Y_\pm := (0, l_\pm), \eta_\pm = h^{-1}y_\pm \in (-1/2, 1/2)\},$$

meet the axis x_1 at the angle $(\pi \mp \alpha)/2$. Here, $h \in (0, 1]$ is a small parameter and $\eta_\pm$ are “rapid” variables in the normal cross-sections of the beams. The junction $\Omega(h)$ consists of two beams $Q^\pm(h)$ and three knots $\mathcal{G}_h^\beta$, $\beta = 0, \pm$ (see Figure 2(a)).

Let $\mathcal{T}_k^\pm(a)$, $k = 0, 1$, be small rectangles

$$\begin{aligned} \mathcal{T}_0^\pm(a) &= \{(y_\pm, z_\pm) \in \mathbf{R}^2 : z_\pm \in (0, a), \eta_\pm \in (-1/2, 1/2)\}, \\ \mathcal{T}_1^\pm(a) &= \{(y_\pm, z_\pm) \in \mathbf{R}^2 : z_\pm \in (l_\pm, l_\pm - a), \eta_\pm \in (-1/2, 1/2)\} \end{aligned} \quad (1.15)$$

and let G_h^β be domains with piecewise smooth boundaries such that

$$\mathcal{T}_1^\pm(h) \subset G_h^\pm \subset B^\pm(2h), \quad \mathcal{T}_0^\pm(h) \subset G_h^0 \subset B^0(2h),$$

where $B^\beta(r)$ is a disk of radius r with center $P^\beta = (P_1^\beta, P_2^\beta)$, $\beta = 0, \pm$, while $(P_1^0, P_2^0) = (0, 0)$ and $(P_1^\pm, P_2^\pm) = (\pm l_\pm \sin(\alpha/2), l_\pm \cos(\alpha/2))$ in the coordinate system (x_1, x_2) . The knots $\mathcal{G}_h^\beta$ are defined by the formulae

$$\mathcal{G}_h^\pm = G_h^\pm \cup \mathcal{T}_1^\pm(3h), \quad \mathcal{G}_h^0 = G_h^0 \cup \mathcal{T}_0^-(3h) \cup \mathcal{T}_0^+(3h). \quad (1.16)$$

Our asymptotic structures do not change when we add or cut-off the knots by a piece with diameter $O(h)$, but the definition (1.16), involving the end parts (1.15) of the beams will become much more convenient in the justification procedure.

1.4 | The elasticity problem

We set

$$\Omega(h) = Q^-(h) \cup Q^+(h) \cup \mathcal{G}_h^+ \cup \mathcal{G}_h^- \cup \mathcal{G}_h^0, \quad (1.17)$$

where $Q^\pm(h) = Q^\pm(h) \setminus (\mathcal{G}_h^\pm \cup \mathcal{G}_h^0)$. We systematically use matrix rather than tensor notation (see [11]) because it is much more suitable to describe the asymptotic ansätze. The displacement vector u is regarded as a column $u = (u_1, u_2)^\top$ in the fixed Cartesian system (x_1, x_2) , where $\top$ stands for transposition. We introduce elastic strain and stress columns

$$\varepsilon(u) = (\varepsilon_{11}, \varepsilon_{22}, 2^{1/2}\varepsilon_{12})^\top, \quad \sigma(u) = (\sigma_{11}, \sigma_{22}, 2^{1/2}\sigma_{12})^\top \quad (1.18)$$

(the factors $2^{1/2}$ are inserted into (1.18) in order to equalize the natural norms of strain columns and tensors) and write the Hooke law in the matrix form $\sigma(u) = A\varepsilon(u)$, where (ε_{ij}) is a strain tensor with the Cartesian components

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right) \quad (1.19)$$

and the 3×3 -matrix A contains the elastic moduli of the material. We consider anisotropic and inhomogeneous materials. We note, however, that in the case of isotropic homogeneous material the matrix has only the following non-zero entries: $A_{11} = A_{22} = \lambda + 2\mu$, $A_{33} = 2\mu$, $A_{12} = A_{21} = \lambda$, where $\lambda \geq 0$ and $\mu > 0$ are the Lamé constants.

According to (1.18) and (1.19), we have $\varepsilon(u) = D(\nabla_x)u$, where

$$D(\nabla_x)^\top = \begin{pmatrix} \partial/\partial x_1 & 0 & 2^{-1/2}\partial/\partial x_2 \\ 0 & \partial/\partial x_2 & 2^{-1/2}\partial/\partial x_1 \end{pmatrix}, \quad \nabla_x = \begin{pmatrix} \partial/\partial x_1 \\ \partial/\partial x_2 \end{pmatrix}. \quad (1.20)$$

Then the equilibrium equations takes the form

$$L(h, x, \nabla_x)u(h, x) := D(-\nabla_x)^\top A(h, x)D(\nabla_x)u(h, x) = f(h, x), \quad x \in \Omega(h). \quad (1.21)$$

Here, $f = (f_1, f_2)^\top$ is a vector of volume forces. We impose the boundary conditions

$$B(h, x, \nabla_x)u(h, x) := D(n)^\top A(h, x)D(\nabla_x)u(h, x) = h^{-1}g(h, x), \quad x \in \partial\Omega(h), \quad (1.22)$$

where n is the unit outward normal vector and g is the traction. Let g differ from zero only on small parts $\gamma_\beta(h) \subset \partial\mathcal{G}_h^\beta \cap \partial\Omega(h)$, $\beta = 0, \pm$ of the boundary $\partial\Omega(h)$, that is, the external surface forces are applied to the knots only while the volume forces, e.g., gravitation, act on the whole junction. We suppose that $g^\beta \in H^1(\gamma_\beta(h))$, where $g^\beta(x) = g(x)$ for $x \in \gamma_\beta(h)$, and specify below properties of g^β and f in (1.21) (Section 2.4).

By a weak solution of the problem (1.21), (1.22) in the Sobolev space $H^1(\Omega(h))^2$ (the superscript 2 indicates the number of the vector function components), we understand $u \in H^1(\Omega(h))^2$ such that the following integral identity is valid:

$$\mathcal{E}(u, v; \Omega(h)) = (f, v)_{\Omega(h)} + \sum_{\beta=0, \pm} (g^\beta, v)_{\gamma_\beta(h)}, \quad \forall v \in H^1(\Omega(h))^2. \quad (1.23)$$

Here and below, $(\cdot, \cdot)_Y$ stands for the natural scalar product in the Lebesgue space $L^2(Y)$ (scalar or vector) and $\mathcal{E}(u, u; \Omega(h)) = (AD(\nabla_x)^\top u, D(\nabla_x)u)_{\Omega(h)}$ is the doubled elastic energy of the junction $\Omega(h)$.

Any rigid motion $\mathbf{d}b$ satisfies the homogeneous ($f = 0, g = 0$) problem (1.23), while $b \in \mathbf{R}^3$ is a column and

$$\mathbf{d}(x) = \begin{pmatrix} 1 & 0 & 2^{-1/2}x_2 \\ 0 & 1 & -2^{-1/2}x_1 \end{pmatrix}.$$

Hence, the compatibility conditions in the problem (1.23) look as follows:

$$\int_{\Omega(h)} \mathbf{d}^\top(x) f(h, x) dx + \int_{\partial\Omega(h)} \mathbf{d}^\top(x) g(h, x) ds_x = 0 \in \mathbf{R}^3. \quad (1.24)$$

The next assertion is well-known.

Lemma 1.1. *Problem (1.23) with the right-hand sides $f \in L^2(\Omega(h))^2$ and $g \in L^2(\partial\Omega(h))^2$ has a solution $u \in H^1(\Omega(h))^2$ if and only if the three compatibility conditions (1.24) are valid. The solution u is defined up to a rigid motion $\mathbf{d}b$. It becomes unique under three orthogonality conditions*

$$\int_{\mathcal{G}_h^0} \mathbf{d}^\top(x) u(h, x) dx = 0 \in \mathbf{R}^3 \quad (1.25)$$

and satisfies the estimate $\|u; H^1(\Omega(h))\| \leq c(h)(\|f; L^2(\Omega(h))\| + \|g; L^2(\partial\Omega(h))\|)$.

The multiplier $c(h)$ in the above-mentioned estimate depends on the small parameter h . We will clarify this dependence in §3 while formulating a particular Korn inequality for the junction $\Omega(h)$.

1.5 | A coordinate transformation

The matrix notation requires a fixed coordinate system. We provide formulas due to the orthogonal coordinate change $(x_1, x_2) \mapsto \mathbf{x}_\pm = \theta_\pm x$. To this end we use the orthogonal matrices

$$\theta_\pm = \begin{pmatrix} \cos(\alpha/2) & \pm \sin(\alpha/2) \\ \mp \sin(\alpha/2) & \cos(\alpha/2) \end{pmatrix}, \quad \Theta_\pm = \begin{pmatrix} \cos^2(\alpha/2) & \sin^2(\alpha/2) & \mp 2^{-1/2} \sin \alpha \\ \sin^2(\alpha/2) & \cos^2(\alpha/2) & \pm 2^{-1/2} \sin \alpha \\ \pm 2^{-1/2} \sin \alpha & \mp 2^{-1/2} \sin \alpha & \cos \alpha \end{pmatrix}.$$

The equality $D(\nabla_x) = \Theta_\pm D(\nabla_{\mathbf{x}_\pm}) \Theta_\pm^\top$ is verified by a direct calculation so that the Equations 1.21 and 1.22 in coordinates $\mathbf{x}_\pm$ take the form

$$\begin{aligned} \mathbf{L}_\pm(\nabla_{\mathbf{x}_\pm}) \mathbf{u}_\pm(h, \mathbf{x}_\pm) &:= D(-\nabla_{\mathbf{x}_\pm})^\top A^\pm D(\nabla_{\mathbf{x}_\pm}) \mathbf{u}_\pm(h, \mathbf{x}_\pm) = \mathbf{f}_\pm, \quad \mathbf{x}_\pm \in \Omega(h), \\ \mathbf{B}_\pm(\nabla_{\mathbf{x}_\pm}) \mathbf{u}_\pm(h, \mathbf{x}_\pm) &:= D(v(\mathbf{x}_\pm))^\top A^\pm D(\nabla_{\mathbf{x}_\pm}) \mathbf{u}_\pm(h, \mathbf{x}_\pm) = \mathbf{g}_\pm, \quad \mathbf{x}_\pm \in \partial\Omega(h), \end{aligned} \quad (1.26)$$

where $\mathbf{u}_\pm = \theta_\pm u$, $A^\pm = \theta_\pm^\top A \theta_\pm$ and $\mathbf{f}_\pm = \theta_\pm f$, $\mathbf{g}_\pm = \theta_\pm g$. Strain and stress columns look as follows: $\boldsymbol{\varepsilon}(\mathbf{u}_\pm) = \Theta_\pm^\top \boldsymbol{\varepsilon}(u)$ and $\boldsymbol{\sigma}(\mathbf{u}_\pm) = A_\pm \boldsymbol{\varepsilon}(\mathbf{u}_\pm)$.

2 | FORMAL ASYMPTOTIC EXPANSIONS

2.1 | First terms

The asymptotic ansatz for the solution of problem (1.21), (1.22) is described in the local coordinates $(y_\pm, z_\pm)$ with the unit vectors $e_1^\pm$ and $e_2^\pm$ as follows:

$$u(h, x) \sim h^{-2} U^{-2}(z_\pm) + h^{-1} U^{-1}(z_\pm, \eta_\pm) + h^0 U^0(z_\pm, \eta_\pm) + h^1 U^1(z_\pm, \eta_\pm) + \dots \quad (2.1)$$

According to the general approach,^[14,15] we set

$$U^{-2}(z_\pm) = w_2^\pm(z_\pm) e_2^\pm + a_1^\pm e_1^\pm. \quad (2.2)$$

The expression (2.2) differs by the addendum $a_1^\pm e_1^\pm$ from the traditional expression used for a single beam. The second term also takes the standard form

$$U^{-1}(z_\pm, \eta_\pm) = (w_1^\pm(z_\pm) - \eta_\pm \partial_z w_2^\pm(z_\pm)) e_1^\pm. \quad (2.3)$$

The coordinate change $(x_1, x_2) \mapsto (\eta_\pm, z_\pm) = (h^{-1} y_\pm, z_\pm)$ leads to the splitting of the differential operator (1.20)

$$D(\nabla_x) = h^{-1} D_\eta + D_z, \quad (2.4)$$

where $D_\eta = D(0, \partial_\eta)$ and $D_z = D(\partial_z, 0)$. In (2.4) and in what follows, we omit the indexes “ $\pm$ ”. According to (2.4),

$$\begin{aligned} \mathbf{L}(h, x, \nabla_x) &= h^{-2} L^0(\partial_\eta) + h^{-1} L^1(\partial_z, \partial_\eta) + h^0 L^2(\partial_z), \\ \mathbf{B}(h, x, \nabla_x) &= h^{-1} B^{0,q}(x, \partial_\eta) + h^0 B^{1,q}(x, \partial_z), \end{aligned} \quad (2.5)$$

where

$$\begin{aligned} L^0 &= -D_\eta^\top A D_\eta, \quad L^1 = -D_\eta^\top A D_z - D_z^\top A D_\eta, \quad L^2 = -D_z^\top A D_z, \\ B^{0,q} &= (-1)^q D(e_2)^\top A D_\eta, \quad B^{1,q} = (-1)^q D(e_2)^\top A D_z, \quad q = 0, 1. \end{aligned} \quad (2.6)$$

Here and below, we suppose that the entries in A are smooth functions in η and $z_\pm$. Notice that the final formulas admit the limit passage due to approximation by smooth moduli of piecewise constant ones, typical for laminated beams.

Let the vector function $(h, x) \mapsto f(h, x)$ depend smoothly on the parameter $h \in [0, h_0]$ and the point x in a neighborhood of $\Omega(h)$. Hence, on the beams $\mathcal{Q}_\pm(h)$ we have the Taylor formula

$$f(h, x) = f^{0\pm}(z_\pm) + \dots, \quad f^{0\pm}(z_\pm) = f(0, 0, z_\pm). \quad (2.7)$$

A small remainder indicated by dots will be specified in Section 3. Notice that in Section 4.1 we will propose quite different assumptions on the exterior loadings $f(h, x)$ and $g(h, x)$ in the problem.

We insert (2.1), (2.5), (2.7) into (1.26), collect coefficients of alike powers of h and arrive at the sequence $(k = -2, -1, 0, 1, 2)$ of one-dimensional problems in beam's cross-sections

$$\begin{aligned} L^0 U^k &= F^k := -L^1 U^{k-1} - L^2 U^{k-2} + \delta_{k,2} f^{0\pm}, \quad \eta \in (-1/2, 1/2), \\ B^{0,q} U^k &= G^{k,q} := -B^{1,q} U^{k-1}, \quad \eta = (-1)^q/2, \quad q = 0, 1, \end{aligned} \quad (2.8)$$

where $\delta_{i,j}$ is the Kronecker symbol and $U^p = 0$ for $p < -2$. Each of these problems gets two compatibility conditions

$$\int_{-1/2}^{1/2} F_i^k(z, \eta) d\eta + \sum_{q=0,1} G_i^{k,q}(z) = 0, \quad i = 1, 2. \quad (2.9)$$

The vectors (2.2) and (2.3) obviously satisfy the Equations 2.8 with $k = -2$ and $k = -1$. Moreover, $D_z U^{-1}(z, \eta) = \mathcal{Y}(\eta) D(\partial_z) w(z)$ with the column $w = (w_1, w_2)^\top$ and

$$\mathcal{Y}(\eta) = \begin{pmatrix} 1 & 0 & 0 \\ -2^{1/2}\eta & 0 & 0 \end{pmatrix}^\top, \quad D(\partial_z) = \begin{pmatrix} \partial_z & 0 \\ 0 & 2^{-1/2}\partial_z^2 \end{pmatrix}. \quad (2.10)$$

By (2.2), (2.6) and (2.8), we have $F^0 = D_\eta^\top A D_z U^{-1}$ and $G^{0,q} = (-1)^q D(e_2)^\top A D_z U^{-1}$. Hence,

$$U^0(z, \eta) = \mathcal{X}(\eta, z) D(\partial_z) w(z) + e_1 W(z), \quad (2.11)$$

where the 2×2 -matrix $\mathcal{X}$ is a solution of the problem

$$-D_\eta^\top A D_\eta \mathcal{X} = D_\eta^\top A \mathcal{Y}, \quad \eta \in (-1/2, 1/2), \quad D(e_2)^\top A D_\eta \mathcal{X} = -D(e_2)^\top A \mathcal{Y}, \quad \eta = \pm 1/2. \quad (2.12)$$

We have added a function W into (2.11) to make the asymptotic ansatz suitable for further use in Section 2.4.

2.2 | Derivation of the resulting system for $w^\pm$

The right-hand sides of the problem (2.8) with $k = 1$ take the form

$$F^1 = D_\eta^\top A D_z U^0 + D_z^\top A (D_\eta U^0 + D_z U^{-1}), \quad G^{1,q} = (-1)^q D(e_2)^\top A D_z U^0.$$

The second ($i = 2$) condition (2.9) is verified:

$$\begin{aligned} (e_2)^\top \left(\int_{-1/2}^{1/2} F^1 d\eta + \sum_{q=0,1} G^{1,q} \right) &= (e_2)^\top \int_{-1/2}^{1/2} D_z^\top A (D_\eta U^0 + D_z U^{-1}) d\eta \\ &= \frac{\partial}{\partial z} \int_{-1/2}^{1/2} (D_\eta(\eta) e_1)^\top A (D_\eta U^0 + D_z U^{-1}) d\eta = -(e_1)^\top \frac{\partial}{\partial z} \left(\int_{-1/2}^{1/2} \eta D_\eta^\top A (D_\eta U^0 \right. \\ &\quad \left. + D_z U^{-1}) d\eta \mp \sum_{\pm} \eta (-1)^q D(e_2)^\top A (D_\eta U^0 + D_z U^{-1}) \Big|_{\eta=\pm 1/2} \right) = 0. \end{aligned} \quad (2.13)$$

In the middle part of (2.13), we use integration by parts and the identity

$$D(e_1) e_2 = D(e_2 \partial_\eta) e_1 \eta. \quad (2.14)$$

The last equality in (2.13) follows from (2.8) with $k = 0, -1$.

In view of $D_\eta U^0 + D_z U^{-1} = (D_\eta \mathcal{X} + \mathcal{Y})D(\partial_z)w$, the compatibility condition (2.9) with $i = 1$ becomes

$$-(e_1)^\top D_z^\top \int_{-1/2}^{1/2} A(D_\eta \mathcal{X}(\eta, z) + \mathcal{Y}(\eta)) d\eta D(\partial_z)w(z) = 0. \quad (2.15)$$

It is nothing but the first equation for w , while the second one emerges from the problem (2.8) with $k = 2$ and

$$F^2 = D_\eta^\top A D_z U^1 + D_z^\top A (D_\eta U^1 + D_z U^0) + f_2^{0\pm}, \quad G^{2,q} = (-1)^q D(e_2)^\top A D_z U^1. \quad (2.16)$$

Namely, the second ($i = 2$) compatibility condition (2.9) converts into

$$\begin{aligned} 0 &= (e_2)^\top \left(\int_{-1/2}^{1/2} F^2 d\eta + \sum_{q=0,1} G^{2,q} \right) = \int_{-1/2}^{1/2} \left(\frac{\partial}{\partial z} (D_\eta(\eta e_1))^\top A (D_\eta U^1 + D_z U^0) + f_2^{0\pm} \right) d\eta \\ &= - \int_{-1/2}^{1/2} \left((e_1)^\top \frac{\partial}{\partial z} \eta D_\eta^\top A (D_\eta U^1 + D_z U^0) - f_2^{0\pm} \right) d\eta \\ &\quad \mp (e_1)^\top \frac{\partial}{\partial z} \sum_{\pm} \eta (-1)^q D(e_2)^\top A (D_\eta U^1 + D_z U^0) \Big|_{\eta=\pm 1/2} \\ &= (e_1)^\top \frac{\partial}{\partial z} \left(\int_{-1/2}^{1/2} \eta D_z^\top A (D_\eta U^0 + D_z U^{-1}) d\eta \right) + F_2^\pm. \end{aligned} \quad (2.17)$$

According to (2.10), the system (2.15), (2.17) turns into

$$\mathbf{M}^\pm(z_\pm, \partial_{z_\pm}) w^\pm(z_\pm) := D(-\partial_{z_\pm})^\top \mathcal{M}^\pm(z_\pm) D(\partial_{z_\pm}) w^\pm(z_\pm) = (0, \mathbf{F}_2^\pm(z_\pm)), \quad z_\pm \in Y_\pm, \quad (2.18)$$

with $\mathbf{F}_2^\pm = f_2^{0\pm}(z)$ and the 3×3 -matrix-functions

$$\mathcal{M}^\pm(z) = \int_{-1/2}^{1/2} (D_\eta \mathcal{X}^\pm(z, \eta) + \mathcal{Y}^\pm(\eta))^\top A^\pm(z, \eta) (D_\eta \mathcal{X}^\pm(z, \eta) + \mathcal{Y}^\pm(\eta)) d\eta. \quad (2.19)$$

The last equality follows from (2.12). The Gram matrix $\mathcal{M}^\pm$ of size 2×2 is symmetric and positive definite.

2.3 | Derivation of the resulting system for $W^\pm$

According to (2.16), the first ($i = 1$) condition (2.9) with $k = 2$ becomes

$$\int_{-1/2}^{1/2} \left((e_1)^\top D_z^\top A (D_\eta U^1 + D_z U^0) + f_1^{0\pm} \right) d\eta = 0. \quad (2.20)$$

Being a solution of the problem (2.12), the matrix $\mathcal{X}$ in (2.11) is defined in up to a constant matrix, i.e.,

$$\mathcal{X}(\eta) = \mathbf{X}(\eta) + e_1 \mathbf{c}, \quad \mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2) \in \mathbf{R}^2 \quad (2.21)$$

and $\mathbf{X}$ is of mean zero $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2)$ with $\mathbf{c}_j \in \mathbf{R}$. Then we write

$$U^1(z, \eta) = \mathbf{X}(\eta) D(\partial_z) \partial_z w(z) + \mathcal{X}(\eta) D(\partial_z) W(z), \quad (2.22)$$

where $\mathbf{X}$ is a solution of the problem

$$\begin{aligned} -D_\eta^\top A D_\eta \mathbf{X} &= D_\eta^\top A D_z \mathcal{X} + D_z^\top A (D_\eta \mathcal{X} + D_z \mathcal{Y}), \quad \eta \in (-1/2, 1/2), \\ D(e_2)^\top A D_\eta \mathbf{X} &= D(e_2)^\top A D_z \mathcal{X}, \quad \eta = \pm 1/2. \end{aligned}$$

Owing to (2.11), (2.21), (2.22), the left-hand side of the compatibility condition (2.20) is the sum of two integrals

$$\begin{aligned} & \int_{-1/2}^{1/2} \left((e_1)^\top D_z^\top A (D_\eta \mathcal{X} + \mathcal{Y}) D (\partial_z) e_1 W + f_1^{0\pm} \right) d\eta \\ & + \int_{-1/2}^{1/2} (e_1)^\top D_1^\top A (D_\eta \mathbf{X} + D_1 (\mathcal{X} + e_1 \mathbf{c})) \partial_z^2 D (\partial_z) w d\eta, \end{aligned} \quad (2.23)$$

with $D_1 = D(1, 0)$. Let $\mathbf{a} := e_1^\top D_1^\top A D_1 e_1 > 0$. We set

$$\mathbf{c} = -\mathbf{a}^{-1} \int_{-1/2}^{1/2} e_1^\top D_1^\top A (D_\eta \mathbf{X} + D_1 \mathcal{X}) d\eta \quad (2.24)$$

to annul the second integral in (2.23). As a result, we reduce the condition (2.20) to the form

$$-\partial_{z_\pm} \mathcal{M}_{11}^\pm \partial_{z_\pm} W^\pm(z_\pm) = f_1^{0\pm}(z_\pm) =: \mathbf{F}_1^\pm(z_\pm). \quad (2.25)$$

In (2.25) we use the definition (2.19) of the matrices $\mathcal{M}^\pm$. We prescribe the boundary conditions

$$\mathcal{M}_{11}^\pm \partial_{z_\pm} W^\pm(l_\pm) = \int_{\gamma_\pm(1)} (e_1^\pm)^\top \mathbf{g}^\pm(\mathbf{y}) d\mathbf{y}_s =: \mathbf{G}_1^{\pm\pm}, \quad \mathcal{M}_{11}^\pm \partial_{z_\pm} W^\pm(0) = \mathbf{H}_\pm. \quad (2.26)$$

In order to satisfy the compatibility conditions, we set $\mathbf{H}_\pm = -\mathbf{F}_1^\pm - \mathbf{G}_1^{\pm\pm}$.

Let $W^\pm$ to be of mean zero, cf. (1.11). Then there exists a unique solution of the problem (2.25), (2.26) of mean zero and the estimates $\sum_\pm \|W^\pm; H^{k+1}(\Upsilon_\pm)\| \leq c_k (\|\mathbf{F}_1^\pm; H^{k-1}(\Upsilon_\pm)\| + |\mathbf{G}_1^{\pm\pm}|)$ with $k = 0, 1, 2, \dots$ hold.

2.4 | Boundary and transmission conditions

Let us define boundary and transmission conditions to close the system (2.18). To this end, we specify properties of the functions g^β in (1.22). Let $\mathbf{y}^\beta$ be the “rapid” variables $\mathbf{y}^\beta = (\mathbf{y}_1^\beta, \mathbf{y}_2^\beta) = (h^{-1}(x_1 - P_1^\beta), h^{-1}(x_2 - P_2^\beta))$ in the nodes $\mathcal{G}_h^\beta$, $\beta = 0, \pm$. We suppose that

$$g^\beta(h, x) = \mathbf{g}^s(\mathbf{y}) + \dots, \quad \mathbf{g}^s \in H^1(\mathcal{G}_1^\pm). \quad (2.27)$$

A small remainder indicated by dots will be specified in §3.

According to [14,15], we impose at the point P^0 the continuity of the rotations (1.7) and the translation

$$a_1^- = \sin \alpha w_2^+(0), \quad a_1^+ = \sin \alpha w_2^-(0). \quad (2.28)$$

Let the space $\mathbf{H}$ consist of vectors

$$\mathbf{v} = (v_1^+, v_2^+, v_1^-, v_2^-)^\top \in H^1(\Upsilon_+) \times H^2(\Upsilon_+) \times H^1(\Upsilon_-) \times H^2(\Upsilon_-),$$

where $v_2^\pm$ satisfy the transmission conditions (1.7) and $v_1^\pm$ is subject to the orthogonality conditions (1.11).

To obtain the integral identity serving for the system (2.18) and to complete with conditions at the points P^0 and $P^\pm$, we multiply the equality (2.18) with a test function $v \in \mathbf{H}$ and integrate by parts to arrive at

$$\begin{aligned} \sum_\pm (\mathcal{M}^\pm D(\partial_z) w^\pm, D(\partial_z) v^\pm)_{\Upsilon_\pm} &= \sum_\pm (D^\top (-\partial_z) \mathcal{M}^\pm D(\partial_z) w^\pm, v^\pm)_{\Upsilon_\pm} \\ &- \left(\mathcal{M}^+ D(\partial_z) w^+(0), \partial_z v_2^+(0) e_2^+ \right) - \left(\mathcal{M}^- D(\partial_z) w^-(0), \partial_z v_2^-(0) e_2^- \right) \\ &- \sum_\pm \left(\left(\mathcal{M}^\pm D(\partial_z) w^\pm(0), v_1^\pm(0) e_1^\pm \right) - \left(\mathcal{M}^\pm \partial_z D(\partial_z) w^\pm(0), v_2^\pm(0) e_2^\pm \right) \right. \\ &- \left(\mathcal{M}^\pm D(\partial_z) w^\pm(l_\pm), v_1^\pm(l_\pm) e_1^\pm \right) + \left(\mathcal{M}^\pm \partial_z D(\partial_z) w^\pm(l_\pm), v_2^\pm(l_\pm) e_2^\pm \right) \\ &\left. - \left(\mathcal{M}^+ D(\partial_z) w^+(l_\pm), \partial_z v_2^+(l_\pm) e_2^+ \right) \right). \end{aligned} \quad (2.29)$$

To eliminate the terms left outside the integral, we impose the boundary equation for the bending moment, the longitudinal and transversal forces, respectively:

$$T^\pm(l_\pm) := (e_2^\pm)^\top \mathcal{M}^\pm D(\partial_z) w^\pm(l_\pm) = 0, \quad (2.30)$$

$$F_1^\pm(l_\pm) := (e_1^\pm)^\top \mathcal{M}^\pm D(\partial_z) w^\pm(l_\pm) = 0, \quad (2.31)$$

$$F_2^\pm(l_\pm) := -(e_2^\pm)^\top \partial_z \mathcal{M}^\pm D(\partial_z) w^\pm(l_\pm) = \mathbf{G}_2^{\pm}, \quad \mathbf{G}_j^{\pm} = \int_{\gamma_\pm(1)} (e_j^\pm)^\top \mathbf{g}^\pm(\mathbf{y}) d\mathbf{y}_s. \quad (2.32)$$

The equilibrium of the knot $\mathcal{G}_h^0$ requires transmission conditions for the bending moment, the longitudinal forces in direction $z_\pm$ and the transversal forces in at direction $y_\pm$, respectively:

$$T^0(0) := (e_2^+)^\top (\mathcal{M}^+ D(\partial_z) w^+(0) + (e_2^-)^\top \mathcal{M}^- D(\partial_z) w^-(0) = 0, \quad (2.33)$$

$$F_1^\pm(0) := (e_1^\pm)^\top \mathcal{M}^\pm D(\partial_z) w^\pm(0) = 0, \quad (2.34)$$

$$F_2^\pm(0) := (e_2^\pm)^\top \partial_z \mathcal{M}^\pm D(\partial_z) w^\pm(0) + \cos \alpha (e_2^\mp)^\top \partial_z \mathcal{M}^\mp D(\partial_z) w^\mp(0) = \pm \sin \alpha \mathbf{H}^\mp - \mathbf{G}_2^{0\pm}, \quad (2.35)$$

$$\mathbf{G}_j^{0\pm} = \int_{\gamma_0(1)} (e_j^\pm)^\top \mathbf{g}^0(\mathbf{y}) d\mathbf{y}_s.$$

By (2.35), we obtain

$$(e_2^\pm)^\top \partial_z \mathcal{M}^\pm D(\partial_z) w^\pm(0) = (\sin \alpha)^{-2} (\cos \alpha (\mathbf{G}_2^{0\mp} \pm \sin \alpha \mathbf{H}^\pm) - \mathbf{G}_2^{0\pm} \pm \sin \alpha \mathbf{H}^\mp). \quad (2.36)$$

2.5 | Solvability of the resultant problem

The weak solution of the resultant problem is a vector $\mathbf{w} = (w_1^+, w_2^+, w_1^-, w_2^-)^\top \in \mathcal{H}$ satisfying the integral identity

$$\mathbf{E}(w, v) := \sum_{\pm} (\mathcal{M} D(\partial_z) w^\pm, D(\partial_z) v^\pm)_{Y_\pm} = \sum_{\pm} \left((\mathbf{F}_2^\pm, v_2^\pm)_{Y_\pm} - v_2^\pm(l_\pm) \mathbf{G}_2^{\pm} + v_2^\pm(0) (\sin \alpha)^{-2} (\cos \alpha (\mathbf{G}_2^{0\mp} \pm \sin \alpha \mathbf{H}^\pm) - \mathbf{G}_2^{0\pm} \pm \sin \alpha \mathbf{H}^\mp) \right) \quad \forall \mathbf{v} = (v_1^+, v_2^+, v_1^-, v_2^-)^\top \in \mathcal{H}.$$

We represent the system (2.18) with the point conditions (1.7), (2.30)–(2.34), (2.36) in matrix form. The twelve point conditions are obtained by applying a 12×4 -matrix $\mathbf{B}$ of differential operators to the column $\mathbf{w}$. Note that in the case of a boundary condition for $w^\pm$ at the points $z_\pm = l_\pm$ and $z_\pm = 0$ we insert zero entries into $\mathbf{B}$ corresponding to $w^\mp$. Then the one-dimensional problem looks as follows:

$$\mathbf{M}(\partial_{z_\pm}) \mathbf{w}(z_\pm) = \mathbf{F}(z_\pm), \quad z_\pm \in (0, l_\pm), \quad \mathbf{B}(\partial_{z_\pm}) \mathbf{w}(z) \Big|_{z_\pm=0, l_\pm} = \mathbf{G}, \quad (2.37)$$

where $\mathbf{F} = (0, \mathbf{F}_2^+, 0, \mathbf{F}_2^-)^\top$, $\mathbf{M}$ is a 4×4 -matrix-operator of the block structure

$$\mathbf{M}(\partial_z) := \begin{pmatrix} \mathbf{M}^+(\partial_z) & \mathbf{O}_2 \\ \mathbf{O}_2 & \mathbf{M}^-(\partial_z) \end{pmatrix},$$

and $\mathbf{O}_2$ is the null 2×2 -matrix. In (2.37) and below we fix $z = z_\pm$ when we deal with $w^\pm$.

Lemma 2.1. *i) The matrix $\mathbf{M}$ is symmetric and positive definite. ii) The homogeneous problem (2.37) has only three linear independent solutions (1.9).*

Proof. Let $\mathcal{Z}^{j,\pm}$ be columns of the 3×2 -matrix $\mathcal{Z}^\pm := D_\eta \mathcal{X}^\pm + \mathcal{Y}^\pm$. The matrix $\mathbf{M}$ is the Gram matrix constructed from four columns $((\mathcal{Z}^{j,+})^\top, 0, 0, 0)^\top, (0, 0, 0, (\mathcal{Z}^{j,-})^\top)^\top$ with the help of the scalar product $(\mathcal{A}\mathcal{U}, \mathcal{V})$ in $L_2(Y_+ \times Y_-)$ where $\mathcal{A} = \text{diag}\{A^+, A^-\}$ is a block-diagonal matrix. In view of the structure of the matrix D_η , see (1.20) and (2.4), columns of the matrix $\mathcal{Z}^\pm$ inherit the linear independence of columns in the matrix $\mathcal{Y}^\pm$. Hence, the statement i) follows from properties of Gram matrices.

By i), a solution $\mathbf{w} = (\mathbf{w}^+, \mathbf{w}^-)^\top$ of the homogeneous system (2.37) satisfies the equalities $\mathcal{D}\mathbf{w}^\pm = 0$. Hence, $\mathbf{w}^\pm = c_1^\pm e_1^\pm + (c_2^\pm + c_0^\pm z_\pm) e_2^\pm$ and we derive ii) from (1.11) and (1.7). According to Lemma 2.1 ii), the Fredholm alternative provides three compatibility conditions in the problem (2.37). $\square$

Lemma 2.2. *The problem (2.37) is solvable.*

Proof. In order to reconstruct the compatibility conditions, one may insert into the Green formula (2.29) the solution $w^\pm$ and vectors from (1.9). As a result, one obtains three conditions

$$\mathbf{F}_2^\pm + (\sin \alpha)^{-2} (\cos \alpha (\mathbf{G}_2^{0\mp} + \sin \alpha \mathbf{H}^\pm) - \mathbf{G}_2^{0\pm} \pm \sin \alpha \mathbf{H}^\mp) - \mathbf{G}_2^{*\pm} = 0, \quad (2.38)$$

$$\sum_{\pm} \left(\int_{\Upsilon_{\pm}} z_{\pm} \mathbf{F}_2^{\pm}(z_{\pm}) dz_{\pm} + l_{\pm} \mathbf{G}_2^{*\pm} \right) = 0. \quad (2.39)$$

We write the compatibility conditions (1.24) in the coordinate system y_-, y_+ and set $h = 0$. Then we obtain formula (2.39) and the following equalities which readily provide (2.38):

$$\mathbf{F}_2^\pm + \cos \alpha \mathbf{F}_2^\mp + \mathbf{G}_2^{*\pm} + \mathbf{G}_2^{0\pm} + \cos \alpha \mathbf{G}_2^{*\mp} \mp \sin \alpha \mathbf{H}^\mp = 0. \quad \square$$

The solution $\mathbf{w}$ becomes unique under the conditions (1.11) and, for example, the three equalities

$$w_2^\pm(0) = 0, \quad \partial_{z_+} w_2^+(0) + \partial_{z_-} w_2^-(0) = 0. \quad (2.40)$$

Proposition 2.3. *Let $k = 0, 1, 2, \dots$ and $\mathbf{F}_2^\pm \in H^{k-2}(\Upsilon_{\pm})$. There exists a solution $\mathbf{w} \in \mathcal{H}$ of the problem (2.18), (1.7), (2.30) – (2.35), (1.11) subject to the normalization conditions (2.40) and the following estimate is valid:*

$$\sum_{\pm} \|w^\pm; H^{k+2}(\Upsilon_{\pm}) \times H^{k+1}(\Upsilon_{\pm})\| \leq c_k \sum_{\pm} \left(\|\mathbf{F}_2^\pm; H^{k-2}(\Upsilon_{\pm})\| + \sum_{j=1,2} \left(|\mathbf{G}_j^{0\pm}| + |\mathbf{G}_j^{*\pm}| \right) \right).$$

Remark 2.4. In the absence of volume forces, the resulting system (1.7), (2.31)–(2.33) admits the explicit solution

$$w_1^\pm = 0, \quad w_2^\pm(z_\pm) = \mathbf{G}_2^{*\pm}(l_\pm z_\pm^2/2 - z_\pm^3/6) + c_0 z_\pm + c_1^\pm, \quad (2.41)$$

where $c_1^\pm$ and c_0 are arbitrary. The problem (2.25), (2.26) is also solvable explicitly: $W^\pm(z_\pm) = \mathbf{G}_1^{*\pm} z_\pm + C^\pm$.

3 | JUSTIFICATION OF THE ASYMPTOTICS

3.1 | Korn inequality

We will justify the constructed asymptotics in the case of free slopes only because clamping a slope crucially simplifies the estimation of the asymptotic remainders.

An asymptotically sharp weighted anisotropic Korn inequality for arbitrary systems of beams is derived in [14]. Let us specify the results of [14] for the junction $\Omega(h)$. We use the partition (1.17) and derive the Korn inequality for each introduced subdomain of $\Omega(h)$. First of all, the orthogonality conditions (1.25) and coordinates stretching assure the estimate

$$|||u; \mathcal{G}_h^0|||_0^2 := (h^{-2} \|u; L^2(\Omega(h))\|^2 + \|\nabla u; L^2(\Omega(h))\|^2) \leq c \mathcal{E}(u, u; \Omega(h)). \quad (3.1)$$

A family of particular lemmas from the article^[14] allow us to transfer individual Korn inequalities from one element of the junction to another. The beams $\mathcal{Q}_\pm(h)$ are attached to the knot $\mathcal{G}_h^0$. By (3.1) and [14, Lemma 2.2], we have

$$\|u; \mathcal{Q}_\pm(h)\|_\pm \leq c \left(|||u; \mathcal{G}_h^0|||_0 + \mathcal{E}(u, u; \Omega(h))^{1/2} \right), \quad (3.2)$$

where $y = y_\pm, z = z_\pm$ for the beams $\mathcal{Q}_\pm(h)$ and

$$\|u; \mathcal{Q}_\pm(h)\|_\pm := \left(\int_{\mathcal{Q}_\pm(h)} \left(\left| \frac{\partial u_z}{\partial z} \right|^2 + \left| \frac{\partial u_y}{\partial y} \right|^2 + h^2 \left(\left| \frac{\partial u_z}{\partial y} \right|^2 + \left| \frac{\partial u_y}{\partial z} \right|^2 \right) + |u_z|^2 + h^2 |u_y|^2 \right) dx \right)^{1/2}. \quad (3.3)$$

The knot $\mathcal{G}_h^\pm$ joins the beam $\mathcal{Q}_\pm(h)$. According to [14, Lemma 2.3] and the estimate (3.2), the inequality

$$|||u; \mathcal{G}_h^\pm|||_\pm \leq c \left(\|u; \mathcal{Q}_\pm(h)\|_\pm + \mathcal{E}(u, u; \Omega(h)) \right)^{1/2} \quad (3.4)$$

holds and involves the weighted anisotropic norm

$$|||u; \mathcal{G}_h^\pm|||_\pm := \left(\int_{\mathcal{G}_h^\pm} \left(\left| \frac{\partial u_z}{\partial z} \right|^2 + \left| \frac{\partial u_y}{\partial y} \right|^2 + h^1 \left(\left| \frac{\partial u_z}{\partial y} \right|^2 + \left| \frac{\partial u_y}{\partial z} \right|^2 \right) + \frac{1}{h} |u_z|^2 + h^1 |u_y|^2 \right) dx \right)^{1/2}. \quad (3.5)$$

We sum up (3.1), (3.2), (3.4) and obtain the Korn inequality whose asymptotic sharpness has been confirmed in [14]:

$$\sum_{\pm} \left(\|u; \mathcal{Q}_\pm(h)\|_\pm + |||u; \mathcal{G}_h^\pm|||_\pm \right) + |||u; \mathcal{G}_h^0|||_0 \leq c \mathcal{E}(u, u; \Omega(h))^{1/2}. \quad (3.6)$$

3.2 | Approximate solution

Let us introduce a partition of unity on $\Omega(h)$. We define the small squares

$$\begin{aligned} S_\pm^1(h) &= \{(x_1, x_2) : |x_{(3\mp 1)/2}| < h/2, l_\pm - 2h < x_{(3\pm 1)/2} < l_\pm - h\} \subset \mathcal{G}_h^\pm, \\ S_\pm^0(h) &= \{(x_1, x_2) : |x_{(3\mp 1)/2}| < h/2, h < x_{(3\pm 1)/2} < 2h\} \subset \mathcal{G}_h^0. \end{aligned}$$

The cut-off functions $X_h^\pm$ depend only on the longitudinal variable $z_\pm$, equal to 1 on $\mathcal{Q}_q(h)$ and 0 on $\mathcal{G}_h^q \setminus S_q^1(h)$ and $\mathcal{G}_h^0 \setminus (S_+^0(h) \cup S_-^0(h))$ while $|\partial_z^p X_h^q(z)| \leq c_p h^{-p}$, $p = 0, 1, 2$. Furthermore, $\chi_h^\pm$ and $\chi_h^{0,\pm}$ are smooth cut-off functions such that $\text{supp } \chi_h^\beta \subset \mathcal{G}_h^\beta$ and $\chi_h^+ + \chi_h^- + \chi_h^{0,+} + \chi_h^{0,-} + X_h^- + X_h^+ \equiv 1$.

We define the approximate solution $\mathbf{U}$ of the problem (1.21), (1.22) by the formula

$$\mathbf{U}(h, x) = \sum_{\pm} X_h^\pm(z) \mathcal{U}_X^\pm(h, x) + \sum_{\beta=0,\pm} \chi_h^\beta(x) \mathcal{U}_G^\beta(h, x), \quad (3.7)$$

where the asymptotic structure on the beams

$$\mathcal{U}_X(h, x) = h^{-2} U^{-2}(z) + h^{-1} U^{-1}(\eta, z) + h^0 U^0(\eta, z) + h^1 U^1(\eta, z) \quad (3.8)$$

is augmented with the asymptotic structures on the knots

$$\begin{aligned} \mathcal{U}_G^\pm(h, x) &= \chi_h^\pm(z) h^{-2} w_2^\pm(l_\pm) + h^{-1} (w_1^\pm(l_\pm) - \tau \partial_z w_2^\pm(l_\pm)), \\ \mathcal{U}_G^0(h, x) &= h^{-2} (a^- e_1^- + a^+ e_1^+) + h^{-1} (e_1^- (w_1^-(0) - \eta_2^- \partial_z w_2^-(0)) + e_1^+ (w_1^+(0) - \eta_2^+ \partial_z w_2^+(0))). \end{aligned} \quad (3.9)$$

The ingredients U^{-2} , U^{-1} , U^0 and $a^\pm$ are defined by (2.3), (2.2), (2.11) and (2.28), respectively. To simplify the notation in (3.8) and below, we omit index “ $\pm$ ”.

In order to specify the representations (2.7) and (2.27) of f and g , we assume that

$$\begin{aligned} f(h, x) &= f^{0\pm}(z_\pm) + \tilde{f}^\pm(h, x), \quad f^{0\pm} \in L^2(\Upsilon_\pm), \quad \|\tilde{f}^\pm; L^2(\Upsilon_\pm)\| \leq Ch, \\ g(h, x) &= \mathbf{g}^s(\mathbf{y}^\beta) + \tilde{\mathbf{g}}^s(\mathbf{y}^\beta), \quad \mathbf{g}^s \in H^1(\partial \mathcal{G}_1^\beta), \quad \|\tilde{\mathbf{g}}^\beta; H^1(\partial \mathcal{G}_1^\beta)\| \leq Ch \end{aligned} \quad (3.10)$$

and introduce the value

$$\mathcal{N} := \sum_{\pm} \left(\|f^{0\pm}; L^2(\Upsilon_\pm)\| + h^{-1} \|\tilde{f}^\pm; L^2(\Upsilon_\pm)\| \right) + \sum_{\beta=0,\pm} \left(\|\mathbf{g}^\beta; H^1(\partial \mathcal{G}_1^\beta)\| + h^{-1} \|\tilde{\mathbf{g}}^\beta; H^1(\partial \mathcal{G}_1^\beta)\| \right). \quad (3.11)$$

Proposition 2.3 and formulas (3.10) assure that $w^\pm \in H^4(\Upsilon_\pm) \times H^3(\Upsilon_\pm)$. These determine all terms in the ansätze (3.8) and (3.9). Moreover, we have the inclusion $U^j \in H^{2-j}(\Upsilon_\pm)^2$ and the estimate

$$\|U^j; H^{2-j}(\Upsilon_\pm)^2\| \leq c \mathcal{N}. \quad (3.12)$$

3.3 | Calculation of discrepancies

Let us consider discrepancies generated by $\mathcal{U}_X^\pm$ in (1.21) and (1.22). First of all, we substitute the sum (3.8) into (1.21). By (2.8), we obtain

$$\begin{aligned} L\mathcal{U}_X^\pm &= h^{-4}L^0U^{-2} + h^{-3}(L^0U^{-1} + L^1U^{-2}) + h^{-2}(L^0U^0 + L^1U^{-1} + L^2U^{-2}) \\ &\quad + h^{-1}(L^0U^1 + L^1U^0 + L^2U^{-1}) + h^0(L^2U^0 + L^1U^1) + h^1L^2U^1 =: \mathbf{F}_\pm. \end{aligned}$$

According to the definition of U^k , we have

$$\mathbf{F}_\pm = h^0(L^1U^1 + L^2U^0) + h^1L^2U^1 \quad \text{and similarly} \quad B\mathcal{U}_X^\pm = h^1B^{1,j}U^1 =: \mathcal{G}_\pm^j, \quad j = 1, 2. \quad (3.13)$$

The difference $\mathcal{R} = u - \mathbf{U}$ satisfies the boundary-value problem

$$\begin{aligned} L\mathcal{R}(h, x) &= f(h, x) - \mathbf{F}_\pm(h, x), \quad x \in \mathcal{Q}_\pm(h), \\ B\mathcal{R}(h, x) &= \mathcal{G}_\pm^j(h, x), \quad x \in \partial\mathcal{Q}_\pm(h), \quad B\mathcal{R}(h, x) = h^{-1}g(h, x), \quad x \in \partial\mathcal{G}_h^\beta(h). \end{aligned} \quad (3.14)$$

The vector $\mathcal{R}$ leaves the discrepancy in the Equations 1.21 on $\mathcal{G}_h^\beta$ the

$$L\mathcal{R}(h, x) = L(X_h^\pm(z_\pm) - 1)\mathcal{U}_X^\pm(h, x) + \mathbf{F}_\pm(h, x) + L(\chi_h^\beta(x))\mathcal{U}_G^\beta(h, x), \quad x \in \mathcal{G}_h^\beta, \quad \beta = 0, \pm, \quad (3.15)$$

with a support in the union of the sets $\overline{S_\pm^0(h)}$ and $\overline{S_\pm^1(h)}$ where the cut-off function is not a constant.

We multiply (3.14) and (3.15) with $\mathcal{R}$ and integrate by parts taking the identity $D(\nabla_x)\mathcal{U}_X^\beta(x) = 0$, $x \in \overline{\mathcal{G}_h^\beta}$, and formulas (1.22), (3.14), (3.15), (3.13) into account. We have

$$\begin{aligned} \mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h)) &= (f, \mathcal{R})_{\Omega(h)} + \sum_{\pm} \left(-(\mathcal{G}_\pm, \mathcal{R})_{\Sigma_\pm} - (\mathbf{F}_\pm, \mathcal{R})_{\mathcal{Q}_\pm(h) \cup S_\pm^0(h) \cup S_\pm^1(h)} \right. \\ &\quad + \left(D(\nabla_x)^\top AD(\nabla_x) \left((X_\pm - 1)\mathcal{U}_X^\pm + \chi_\pm \mathcal{U}_X^\pm \right), \mathcal{R} \right)_{S_\pm^1(h)} \\ &\quad + \left(D(\nabla_x)^\top AD(\nabla_x) \left((X_\pm - 1)\mathcal{U}_X^\pm + \chi_0 \mathcal{U}_X^0, \mathcal{R} \right) \right)_{S_\pm^0(h)} \\ &\quad - \left(D(\nu^{1\pm})^\top AD(\nabla_x) \left((X_\pm - 1)\mathcal{U}_X^\pm + \chi_\pm \mathcal{U}_X^\pm \right), \mathcal{R} \right)_{\Sigma_\pm \cap \partial S_\pm^1(h)} \\ &\quad \left. - \left(D(\nu^{0\pm})^\top AD(\nabla_x) \left((X_\pm - 1)\mathcal{U}_X^\pm + \chi_0 \mathcal{U}_X^0 \right), \mathcal{R} \right)_{\Sigma_\pm \cap \partial S_\pm^0(h)} \right) - h^{-1} \sum_{\beta=0,\pm} (g^\beta, \mathcal{R})_{\gamma_s(h)}. \end{aligned} \quad (3.16)$$

Here, $\nu^{k\pm}$ is the external normal vector on $\Sigma_\pm^k$ and $\Sigma_\pm = \Sigma_\pm^1 \cup \Sigma_\pm^2$. Integrating by parts in (3.16), we get rid of integrals along the lateral sides of the beams. As a result, we obtain

$$\mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h)) = \sum_{\pm} \left(J_X^\pm + \tilde{I}^\pm + I_0^\pm + I_{01}^\pm + h^1 I_1^\pm - \sum_{k=0,1} (-1)^k J_{k\gamma}^\pm \right) + h^{-1} \sum_{\beta=0,\pm} I_\gamma^\beta \quad (3.17)$$

where

$$\begin{aligned} J_X^\pm &= \left(AD(\nabla_x) \left((1 - X_\pm)\mathcal{U}_X^\pm - \chi_\pm \mathcal{U}_X^\pm \right), D(\nabla_x)\mathcal{R} \right)_{S_\pm^1(h)} \\ &\quad + \left(AD(\nabla_x) \left((1 - X_\pm)\mathcal{U}_X^\pm - \chi_0 \mathcal{U}_X^0 \right), D(\nabla_x)\mathcal{R} \right)_{S_\pm^0(h)}, \\ I_{01}^\pm &= h \left(D(\nu^{k\pm})^\top AD_z U^1, \mathcal{R} \right)_{\Sigma_\pm} - \left(D_\eta AD_z U^1, \mathcal{R} \right)_{\mathcal{Q}_\pm(h)}, \\ I_0^\pm &= (f^{0\pm}, \mathcal{R})_{\mathcal{Q}_\pm(h)} + (D_z^\top A(D_\eta U^1 + D_z U^0), \mathcal{R})_{\mathcal{Q}_\pm(h)}, \quad I_1^\pm = (-D_z^\top AD_z U^1, \mathcal{R})_{\mathcal{Q}_\pm(h)}, \\ \tilde{I}^\pm &= (\tilde{f}^\pm, \mathcal{R})_{\mathcal{Q}_\pm(h)}, \quad J_{k\gamma}^\pm = (D(e_1^\pm)^\top AD(\nabla_x)\mathcal{U}_X^\pm, \mathcal{R})_{\gamma_\pm^k(h)}, \quad I_\gamma^\beta = -(g^\beta, \mathcal{R})_{\gamma_s(h)} \end{aligned} \quad (3.18)$$

and

$$\begin{aligned}\gamma_{\mp}^1(h) &= \{(x_1, x_2) : x_{(3\mp 1)/2} = l_{\mp} - 2h, |x_{(3\pm 1)/2}| < h/2\}, \\ \gamma_{-}^0(h) &= \{(x_1, x_2) : x_1 = 2h, |x_2| < h/2\}, \quad \gamma_{+}^0(h) = \{(x_1, x_2) : x_1 = 2h, |x_2| < h/2\}.\end{aligned}$$

The immediate objective becomes to estimate integrals in (3.18) by the quantity $c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}$.

3.4 | Estimating the discrepancy on the beams

By the definition (3.3) and the Korn inequality (3.2), we have

$$|\tilde{I}^{\pm}| \leq ch \|\tilde{f}^{\pm}; L^2(Q_{\pm}(h))\| \|\mathcal{R}; L^2(Q_{\pm}(h))\| \leq c\mathcal{N} \|\mathcal{R}; Q_{\pm}(h)\|_{\pm} \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}.$$

The estimate (3.12) implies

$$\begin{aligned}h|I_1^{\pm}| &\leq ch \|D_z^{\top} A D_z U^1; L^2(Q_{\pm}(h))\| \|\mathcal{R}; L^2(Q_{\pm}(h))\| \leq \\ &\leq c\mathcal{N} \|\mathcal{R}; L^2(Q_{\pm}(h))\| \leq c\mathcal{N} \|\mathcal{R}; Q_{\pm}(h)\|_{\pm} \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}.\end{aligned}$$

Integrating by parts yields $I_{01}^{\pm} = h(AD_z U^1, D_y \mathcal{R})_{Q_{\pm}(h)}$ where $y = h\eta$ and $D_y = D(e_2^{\pm} \partial_y)$. By (3.12) and (3.2), we derive the estimate

$$|I_{01}^{\pm}| \leq c\mathcal{N} \|\mathcal{R}; Q_{\pm}(h)\|_{\pm} \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}.$$

Let us consider $I_0^{\pm}$. The next lemma is known.

Lemma 3.1 (see [14, Lemma 4.1; 12, Lemma 3.4]). *For $R \in H^1(Q_{\pm}(h))^2$, there holds*

$$\begin{aligned}\|R_1; L^2(Q_{\pm}(h))\|^2 + \|R_2 - \langle R_2 \rangle e_2^{\pm}; L^2(Q_{\pm}(h))\|^2 &\leq c\mathcal{E}(R; R; Q_{\pm}(h)), \\ \langle R_2 \rangle(z) &= \frac{1}{h} \int_{-h/2}^{h/2} R_2(y, z) dy.\end{aligned}\tag{3.19}$$

The second line of the system (2.18) gives us

$$0 = h \left(-D_z A (D_z U^0 + D_{\eta} U^1), \langle R_2 \rangle \right)_{Q_{\pm}(h)} - h \left(f_2^{0\pm}, \langle R_2 \rangle \right)_{Y_{\pm}}$$

so that the inequalities (3.19) and (3.12) provide

$$|I_0^{\pm}| \leq c\mathcal{N} \|\mathcal{R}; Q_{\pm}(h)\|_{\pm} \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}.$$

3.5 | Estimating the discrepancy on the small squares

Let us consider the integrals $J_X^{\pm}$. By formulas (3.9) and the obvious equalities $\chi_h^{\pm} + \chi_h^{0,\pm} + X_h^{\pm} = 1$, $\nabla \chi_h^{\pm} + \nabla \chi_h^{0,\pm} + \nabla X_h^{\pm} = 0$, we conclude that

$$J_X^{\pm} = \sum_{\pm} \sum_{q=0,1} \left(AD(\nabla)(1 - X_h^{\pm}) \widetilde{U}^{q\pm}, D(\nabla) \mathcal{R} \right)_{S_{\pm}^q},$$

where the functions $\widetilde{U}^{q\pm}$ look as follows

$$\begin{aligned}\widetilde{U}^1(z) &= h^{-2} e_2 (w_2(z) - w_2(l) - (l - z) \partial_z w_2(l)) + h^{-1} (e_1(w_1(z) \\ &\quad - w_1(l) - \eta(\partial_z w_2(z) - \partial_z w_2(l))) + h^0 U^0(l, \eta) + h^1 U^1(l, \eta), \\ \widetilde{U}^0(z) &= h^{-2} e_2 (w_2(z) - w_2(0) - z \partial_z w_2(0)) + h^{-1} (e_1(w_1(z) \\ &\quad - w_1(0) - \eta(\partial_z w_2(z) - \partial_z w_2(0))) + h^0 U^0(0, \eta) + h^1 U^1(0, \eta),\end{aligned}\tag{3.20}$$

where we omitted the index “ $\pm$ ” for the sake of simplicity. The necessary estimate of $J_X^\pm$ follows from the variants of Hardy's inequality (cf. [13, Lemma A.1]).

Lemma 3.2. *If $\mathcal{V} \in H^1 \circ (Q_\pm(h))$ and $\mathcal{W} \in H^1(Q_\pm(h))$, then*

$$\|\mathcal{V}; L^2(S_\pm^q)\| \leq ch^2 \|\nabla \mathcal{V}; L^2(S_\pm^q)\|, \quad \|\mathcal{W}; L^2(S_\pm^q)\| \leq ch \|\mathcal{W}; H^1(Q_\pm(h))\|. \quad (3.21)$$

We apply the inequalities (3.21) to terms in (3.20) and conclude that

$$\left| (\widetilde{\mathcal{U}}^{q\pm}, D(\nabla)\mathcal{R})_{S_\pm^q} \right| \leq ch^{1/2} h^{1/2} \mathcal{N} \|D(\nabla)\mathcal{R}; L^2(S_\pm^q)\| \leq ch \|\mathcal{R}; H^1(Q_\pm(h))\|. \quad (3.22)$$

The additional factor $h^{1/2}$ came into (3.22) as a result of integration along the mid-line of the square $S_\pm^q$. Since $\|\mathcal{R}; L^2(S_\pm^q)\| \leq ch^{-1} \|\mathcal{R}; Q_\pm(h)\|_\pm$, we obtain

$$|\widetilde{\mathcal{U}}^{q\pm}| \leq ch^0 \mathcal{N} \|\mathcal{R}; Q_\pm(h)\|_\pm \leq c \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}.$$

3.6 | Estimating the discrepancy on surfaces of the knots

Let us consider the last two sums in (3.17). By formula (3.8), the definition of the terms U^k and the decomposition (2.4) of the operator $D(\nabla_x)$, we have

$$J_{k\gamma}^\pm = h^{-1} I_{k\gamma}^\pm + h^0 J_{k\gamma}^\pm + h^1 \mathcal{K}_{k\gamma}^\pm, \quad (3.23)$$

$$\begin{aligned} I_{k\gamma}^\pm &= (D(e_1^\pm)^\top A(D_\eta U^0 + D_z U^{-1}), \mathcal{R})_{\gamma_\pm^k(h)} = (D(e_1^\pm)^\top A(D_\eta \mathcal{X}^\pm + \mathcal{Y}^\pm) D(\partial_z) w^\pm, \mathcal{R})_{\gamma_\pm^k(h)}, \\ J_{k\gamma}^\pm &= (D(e_1^\pm)^\top A(D_z U^0 + D_\eta U^1), \mathcal{R})_{\gamma_\pm^k(h)}, \quad \mathcal{K}_{k\gamma}^\pm = (D(e_1^\pm)^\top A D_z U^1, \mathcal{R})_{\gamma_\pm^k(h)}. \end{aligned}$$

The first estimate

$$h |\mathcal{K}_{k\gamma}^\pm| \leq ch \|AD_z U^1; L^2(\gamma_\pm^k(h))\| \|D(\nabla_x) \mathcal{R}; L^2(\gamma_\pm^k(h))\| \leq ch^{1/2} \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h))^{1/2}$$

follows from the standard trace inequality which can be derived by stretching coordinates, namely

$$h^1 \|\mathcal{V}; L^2(\gamma_\pm^k(h))\|^2 \leq c (\|\mathcal{V}; L^2(S_\pm^k(h))\|^2 + h^2 \|\nabla_x \mathcal{V}; L^2(S_\pm^k(h))\|^2). \quad (3.24)$$

Next we write representations

$$\mathcal{R}(h, x(y_\pm, z_\pm)) = \mathcal{R}_\pm^{k\pm}(h, x(y_\pm, z_\pm)) + \mathbf{d}_\pm^k(y_\pm, z_\pm) \mathbf{r}^{k\pm}(h), \quad x \in S_\pm^k(h), k = 0, 1, \quad (3.25)$$

where

$$\begin{aligned} \mathbf{d}_\pm^0(y_\pm, z_\pm) &= \mathbf{d}(y_\pm, z_\pm), \quad \mathbf{d}_\pm^1(y_\pm, z_\pm) = \begin{pmatrix} 1 & 0 & 2^{-1/2} y_\pm \\ 0 & 1 & 2^{-1/2} (l_\pm - z_\pm) \end{pmatrix}, \quad \int_{S_\pm^k(h)} \mathcal{R}_\pm^{k\pm}(h, x) dx = 0, \\ \mathbf{r}^{k\pm}(h) &= \left(\int_{S_\pm^k(h)} \mathbf{d}_\pm^k(y_\pm, z_\pm) \mathbf{d}_\pm^k(y_\pm, z_\pm)^\top dy_\pm dz_\pm \right)^{-1} \int_{S_\pm^k(h)} \mathbf{d}_\pm^k(y_\pm, z_\pm)^\top \mathcal{R}(h, x(y_\pm, z_\pm)) dy_\pm dz_\pm. \end{aligned} \quad (3.26)$$

Using (3.25) and (3.26), we obtain

$$\begin{aligned} h^{-1} I_{k\gamma}^\pm &= h^{-1} \sum_{q=0}^3 I_q^{k\pm}, \quad I_q^{k\pm} := \left(D(e_1^\pm)^\top A(D_\eta \mathcal{X}^\pm + \mathcal{Y}^\pm) D(\partial_z) w^\pm, R_q^{k\pm} \right)_{\gamma_\pm^k(h)}, \\ R_0^{k\pm} &= \mathcal{R}_\pm^{k\pm}, \quad R_1^{k\pm} = e_1^\pm \mathbf{r}_1^{k\pm}, \quad R_2^{k\pm} = e_2^\pm \mathbf{r}_2^{k\pm}, \quad R_3^{k\pm} = 2^{-1/2} (x_2, -x_1)^\top \mathbf{r}_3^{k\pm}. \end{aligned} \quad (3.27)$$

Taking the identity $e_1^\top D^\top(e_1) = e_1^\top \mathcal{Y}(\eta)^\top$, formula (2.19) for the matrices $\mathbf{M}^\pm$ and the conditions (2.31), (2.34) into account, we find that $I_1^{k\pm} = 0$. By (2.14) and similarly to (2.13), we obtain

$$I_2^{k\pm} = \left(e_1^\top (D(e_2^\pm \partial_{\eta_\pm}) \eta_\pm)^\top A(D_\eta \mathcal{X}^\pm + \mathcal{Y}^\pm) D(\partial_z) w^\pm, \mathbf{r}_2^{k\pm} \right)_{\gamma_\pm^k(h)} = 0.$$

The identities

$$h^{-1} 2^{-1/2} x_2 e_1^\top D^\top(e_1) = (e_2^-)^\top \mathcal{Y}^{-\top}(\eta^-), \quad h^{-1} 2^{-1/2} x_1 e_2^\top D^\top(e_1) = (e_2^+)^\top \mathcal{Y}^{+\top}(\eta^+),$$

formula (2.19), the boundary conditions (2.30) and the transmission condition (2.33) provide the equalities $I_3^{1\pm} = 0$ and $I_3^{0\pm} + I_3^{0\mp} = 0$.

We estimate the terms $I_0^{k\pm}$ using the trace inequality (3.24) and the Korn inequality (3.1) supported by the orthogonality condition in (3.26). In this way we arrive at the relation

$$\begin{aligned} h^{-1} |I_0^{k\pm}| &\leq ch^{-1} h^{1/2} \mathcal{N} \|\mathcal{R}_\pm^{k\pm}; L^2(\gamma_\pm^k(h))\| \leq ch^{-1} h^{1/2} h^{-1/2} \mathcal{N} (\|\mathcal{R}_\pm^{k\pm}; L^2(S_\pm^k(h))\|^2 \\ &\quad + h^2 \|\nabla_x \mathcal{R}_\pm^{k\pm}; L^2(S_\pm^k(h))\|^2)^{1/2} = c \mathcal{N} \|(\mathcal{R}_\pm^{k\pm}; S_\pm^k(h))\|_0 \\ &\leq c \mathcal{N} \mathcal{E}(\mathcal{R}_\pm^{k\pm}, \mathcal{R}_\pm^{k\pm}; S_\pm^k(h)) = c \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; S_\pm^k(h))^{1/2}. \end{aligned} \quad (3.28)$$

It remains to process the integrals $J_{k\gamma}^\pm$ in (3.23). We consider them together with the terms I_γ^β from (3.17). On the knots $\mathcal{G}_h^\pm$ we apply the representation (3.25) and conclude that

$$\begin{aligned} J_{1\gamma}^\pm &= \sum_{q=0}^3 J_q^{1\pm}, \quad J_q^{1\pm} := \left(D(e_1^\pm)^\top A(D_z U^0 + D_\eta U^1), R_q^{1\pm} \right)_{\gamma_\pm^1(h)}, \\ h^{-1} I_\gamma^\pm &= \sum_{q=0}^3 I_{q\gamma}^{1\pm}, \quad I_{q\gamma}^{1\pm} := \left(\mathbf{g}^\pm + \tilde{\mathbf{g}}^\pm, R_q^{1\pm} \right)_{\gamma_\pm^1(1)}. \end{aligned} \quad (3.29)$$

Analogously to (3.28) we derive the estimate

$$h^0 (|J_0^{1\pm}| + |I_{0\gamma}^{1\pm}|) \leq ch^1 \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; S_\pm^k(h)).$$

The third column of the matrix $\mathbf{d}_\pm^k$ gets order h^1 on $S_\pm^k(h)$. Hence,

$$h^0 (|J_3^{1\pm}| + |I_{3\gamma}^{1\pm}|) \leq ch^1 \mathcal{N} \|\mathcal{R}_\pm^{1\pm}; L^2(\mathcal{G}_h^\pm)\| \leq c \mathcal{N} \|\mathcal{R}; \mathcal{G}_h^\pm\|_\pm \leq c \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{G}_h^\pm). \quad (3.30)$$

Owing to (2.11), (2.21), (2.22), (2.24), we have

$$J_{1\pm}^{1\pm} = -\mathcal{M}_{11}^\pm \partial_{z_\pm} W^\pm(l_\pm). \quad (3.31)$$

Moreover, (2.26) yields

$$J_1^{1\pm} + I_{1\gamma}^{1\pm} = - \left((e_1^\pm)^\top (\mathbf{g}^\pm + \tilde{\mathbf{g}}^\pm), \mathbf{r}_1^{1\pm} \right)_{\gamma_\pm(1)} + \left(e_1^\pm \mathbf{G}_1^{\pm\pm}, \mathbf{r}_1^{1\pm} \right)_{\gamma_\pm(1)}.$$

The same chain of equalities as in (2.17) implies

$$J_2^{2\pm} = - (e_2^\pm)^\top \partial_z \mathcal{M}^\pm D(\partial_z) w^\pm(l_\pm). \quad (3.32)$$

Taking the boundary condition (2.32) into account, we write

$$J_2^{1\pm} + I_{2\gamma}^{1\pm} = - \left(e_2^\pm (\mathbf{g}^\pm + \tilde{\mathbf{g}}^\pm), \mathbf{r}_2^{1\pm} \right)_{\gamma_\pm(1)} + \left(e_2^\pm \mathbf{G}_1^{\pm\pm}, \mathbf{r}_2^{1\pm} \right)_{\gamma_\pm(1)}.$$

The above formulas, the inequalities

$$\|(e_j^\pm)^\top \mathbf{g}^\pm - (e_j^\pm)^\top \mathbf{G}_j^{\pm\pm}; L^2(\gamma_\pm(1))\| \leq h \|\mathbf{g}^\pm; H^1(\gamma_\pm(h))\|, \quad j = 1, 2 \quad (3.33)$$

and (3.27), (3.24), (3.10) provide $|J_j^{1\pm} + I_{j\gamma}^{1\pm}| \leq ch^{1/2} \mathcal{N} \|R_2^{1\pm}; H^1(S_\pm^1(h))\|$, $j = 1, 2$. Hence, the formulas (3.4) and (3.6) give

$$|J_j^{1\pm} + I_{j\gamma}^{1\pm}| \leq ch^0 \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; \Omega(h)), \quad j = 1, 2.$$

In order to deal with the integrals $J_0^{q\pm}$ and I_γ^0 we introduce the basis $e_2^+ = (-\cos \alpha/2, \sin \alpha/2)^\top$, $e_2^- = (-\cos \alpha/2, -\sin \alpha/2)^\top$ and the corresponding 2×3 -matrix $\mathbf{d}^\times(x) = (e_2^+, e_2^-, y_- e_2^+ - y_+ e_2^-)$. The difference $\mathcal{R} = u - \mathbf{U}$ admits the representation

$$\mathcal{R}(h, x) = \mathcal{R}_\perp^\times(h, x) + \mathbf{d}^\times(x) \mathbf{r}^\times(h) \quad (3.34)$$

where

$$\mathbf{r}^\times(h, x) = (\mathbf{D}^\times)^{-1} \int_{\mathcal{G}_h^0} \mathbf{d}^\times(x)^\top \mathcal{R}(h, x) dx, \quad \int_{\mathcal{G}_h^0} \mathcal{R}_\perp^\times(h, x) dx = 0, \quad \mathbf{D}^\times = \int_{\mathcal{G}_h^0} \mathbf{d}^\times(x) (\mathbf{d}^\times(x))^\top dx.$$

Using (3.34), we decompose boundary integrals on $\mathcal{G}_h^0$ in the same way as (3.29), namely

$$\begin{aligned} \mathcal{J}_{0\gamma}^\pm &= \sum_{q=0}^3 J_q^\pm, \quad J_q^\pm := \left(D(e_1^\pm)^\top A(D_z U^0 + D_\eta U^1), R_q^\times \right)_{\gamma_\pm^0(1)}, \quad R_0^\times = \mathcal{R}_\perp^\times, \quad R_1^\times = e_2^+ \mathbf{r}_1^\times, \\ R_2^\times &= e_2^- \mathbf{r}_2^\times, \quad R_3^\times = 2^{-1/2} (y_- e_2^+ - y_+ e_2^-) \mathbf{r}_3^\times, \quad h^{-1} I_\gamma^0 = \sum_{q=0}^3 I_{q\gamma}^\times, \quad I_{q\gamma}^\times := \left(\mathbf{g}^0 + \widetilde{\mathbf{g}}^0, R_q^0 \right)_{\gamma_0(1)}. \end{aligned}$$

The inequalities

$$h^0 \left(\sum_{\pm} |J_q^\pm| + |I_{q\gamma}^\times| \right) \leq c \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{G}_h^0), \quad q = 0, 3, \quad (3.35)$$

are derived quite similar to the estimates (3.28) and (3.30) in the cases $q = 0$ and $q = 3$ respectively. In order to estimate the integrals $J_q^{0\pm}$, $q = 1, 2$, we write the vector functions $\mathbf{g}^0$ and $\widetilde{\mathbf{g}}^0$ in the form

$$\mathbf{g} = \mathbf{g}^{0-} e_2^- + \mathbf{g}^{0+} e_2^+, \quad \widetilde{\mathbf{g}}^0 = \widetilde{\mathbf{g}}^{0-} e_2^- + \widetilde{\mathbf{g}}^{0+} e_2^+.$$

Let $I_{1\gamma}^\times := ((\mathbf{g}^{0+} + \widetilde{\mathbf{g}}^{0+}) e_2^+, \mathbf{r}_1^\times)$, $I_{2\gamma}^\times := ((\mathbf{g}^{0-} + \widetilde{\mathbf{g}}^{0-}) e_2^-, \mathbf{r}_2^\times)$. We have

$$J_1^\pm = ((e_2^\pm)^\top D(e_1^\pm)^\top A(D_z U^0 + D_\eta U^1), \mathbf{r}_1^\times), \quad J_2^\pm = ((e_2^\pm)^\top D(e_1^\pm)^\top A(D_z U^0 + D_\eta U^1), \mathbf{r}_2^\times). \quad (3.36)$$

We represent left-hand sides of (3.36) analogously to (3.31) and (3.32). According to (2.35) we obtain

$$\begin{aligned} \sum_{\pm} J_1^\pm + I_{1\gamma}^\times &= -((e_2^+)^\top (\mathbf{g}^{0+} + \widetilde{\mathbf{g}}^{0+}), \mathbf{r}_1^\times)_{\gamma_\pm(1)} + ((e_2^+)^\top \mathbf{G}_2^{\bullet+}, \mathbf{r}_1^\times)_{\gamma_\pm(1)}, \\ \sum_{\pm} J_2^\pm + I_{2\gamma}^\times &= -((e_2^-)^\top (\mathbf{g}^{0+} + \widetilde{\mathbf{g}}^{0+}), \mathbf{r}_2^\times)_{\gamma_\pm(1)} + ((e_2^-)^\top \mathbf{G}_2^{\bullet-}, \mathbf{r}_2^\times)_{\gamma_\pm(1)}. \end{aligned}$$

By the Friedrichs inequality (3.33) and the estimates (3.10), (3.24), (3.1), (3.6), we deduce (3.35) with $q = 1, 2$.

3.7 | The final theorem on the asymptotics

We have estimated all terms on the right-hand side of (3.22) by the quantity $c \mathcal{N}(f) \mathcal{E}(\mathcal{R}; \Omega_h)^{1/2}$. This give us the following error estimate.

Theorem 3.3. *Let the right-hand sides of the problem (1.21), (1.22) satisfy the assumptions (3.10). Then the solution u of this problem and the ansatz (3.7) are related by the inequality*

$$\mathcal{E}(u - \mathbf{U}; \Omega_h) \leq C h^0 \mathcal{N},$$

where $\mathcal{N}$ is the value (3.11) and the constant C is independent of the right-hand sides and the parameter $h \in (0, h_0]$.

Due to the Korn inequality (3.6), Theorem 3.3 provides an estimate for the accuracy of one-dimensional asymptotic model of the junction $\Omega(h)$. By the standard scheme (see e.g. [18, §5.4]), we remove unnecessary “extra” terms from (3.7) and obtain the estimates

$$\begin{aligned} \|u_2^{h\pm} - h^{-2}w_2^\pm; L^2(Q^\pm(h))\| &\leq ch^{1/2}\mathcal{N}, \quad \|u_1^{h\pm} - h^{-2}a_1^\pm - h^{-1}w_1^\pm; L^2(Q^\pm(h))\| \leq ch^{1/2}\mathcal{N}, \\ \|\sigma^\pm(u^{h\pm}) - A(D\mathcal{X}^\pm + \mathcal{Y}^\pm)D(\partial_z)w^\pm; L^2(Q^\pm(h))\| &\leq ch^{1/2}\mathcal{N}. \end{aligned}$$

It is easy to verify the asymptotic accuracy of these estimates, and they do not differ from those in [16], for single beams.

4 | FINAL COMMENTS AND CONCLUSIONS

4.1 | Comparing different ansätze

The matrix $\mathcal{M}^\pm(z_\pm)$ in (2.19) gets the integral representation (see, e.g., [18, §4.2] and [12])

$$\begin{aligned} \mathcal{M}^\pm(z) &= \int_{-1/2}^{1/2} \begin{pmatrix} 1 & -2^{1/2}\eta \\ -2^{1/2}\eta & 2\eta^2 \end{pmatrix} \mathbf{a}_\pm(\eta) d\eta, \\ \mathbf{a}_\pm(\eta, z) &= A_{11}^\pm(\eta, z) - A_{1*}^\pm(\eta, z)A_{**}^\pm(\eta, z)^{-1}A_{*1}^\pm(\eta, z) > 0 \end{aligned} \quad (4.1)$$

where $A_{11}^\pm$ is the upper left entry and $A_{**}^\pm$ is the lower right 2×2 -block of the matrix $A^\pm$. Assuming that $A^\pm$ and therefore $\mathbf{a}_\pm$ are independent of the longitudinal variable $z_\pm$, we set

$$\rho_\pm = \left(\int_{-1/2}^{1/2} \mathbf{a}_\pm(\eta) d\eta \right)^{-1} \int_{-1/2}^{1/2} \eta \mathbf{a}_\pm(\eta) d\eta$$

and perform the coordinate change

$$\eta_\pm \mapsto \eta'_\pm = \eta_\pm + \rho_\pm \text{ or } y \mapsto y'_\pm + h\rho_\pm$$

which makes the matrix (4.1) diagonal. In other words, an appropriate choice of the mid-lines in the beams $Q_\pm(h)$ of the homogeneous junction (1.17) splits the limit system (2.18) into a couple of independent ordinary differential equations of second and fourth orders. Notice that in the case of constant elastic moduli in $A^\pm$ the matrices (4.1) takes the diagonal form $\mathcal{M}^\pm = \text{diag}\{\mathcal{M}_{11}^\pm, \mathcal{M}_{22}^\pm\} = \mathbf{a}_\pm \text{diag}\{1, 1/24\}$ and splitting occurs automatically. Under the above-mentioned disintegration of the limit system, one may change the general asymptotic ansatz and, equalizing orders of the longitudinal and transverse displacement in the beams, in order to compose the asymptotic ansatz for the solution of the problem (1.21), (1.22) with the main term

$$h^{-2}w_2^{0,\pm}(z_\pm)e_1^\pm + h^{-2}w_1^{0,\pm}(z_\pm)e_2^\pm \quad (4.2)$$

(compare (1.1) and (1.2)). First of all, we accept the previous assumptions (2.7), (2.27) and (3.10) on the exterior loadings. Repeating the asymptotic procedure in Section 2, we derive the ordinary differential equations

$$\mathcal{M}_{11}^\pm \partial_{z_\pm}^2 w_1^{0,\pm} = 0, \quad \mathcal{M}_{22}^\pm \partial_{z_\pm}^4 w_2^{0,\pm} = \mathbf{F}_2^\pm, \quad z_\pm \in \Upsilon_\pm, \quad (4.3)$$

which are supplied with the boundary conditions (2.30)–(2.32). However, the accepted change of the ansatz leads to the new continuity conditions at the junction point $z_\pm = 0$

$$w_1^{0,\pm}(0) = \sin \alpha w_2^{0,\mp}(0). \quad (4.4)$$

According to the Green formula (2.29), the appearance of the two stable transmissions in addition to (1.7) reduces to three the number of the intrinsic transmission conditions which now involve the relations (2.33) and (2.35).

The problem obtained involves three compatibility conditions (2.38), (2.39) and the corresponding homogeneous problem has three linear independent solutions $c_1^\pm e_1^\pm + (c_2^\pm + c_0^\pm z^\pm)e_2^\pm$. Hence, the problem with the modified transmission conditions gains

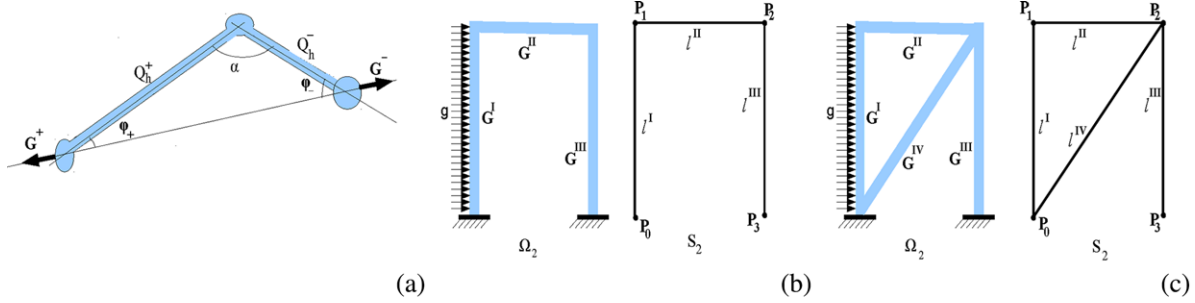


FIGURE 3 (a) The optimization problem. Junctions (b) Ω_1 with its skeleton S_1 and (c) Ω_2 with its skeleton S_2

the same general properties as the problem (2.18), (1.7), (2.30)–(2.35), (1.11) derived in Section 2. Moreover, it is straightforward to verify that a solution of the problem with the transmission conditions (4.4), (1.7) and (2.33), (2.35), (4.4) looks as follows:

$$w_2^{0,\pm}(z^\pm) = w_2^\pm(z^\pm), \quad w_1^{0,\pm}(z^\pm) = \text{const.} \quad (4.5)$$

In order words, under the above-introduced assumption on the elastic material and external loading, the ansätze with the main terms (1.1) and (4.2) are equivalent.

Noting that the last formula in (4.5) is due to the homogeneity of the first equation in (4.3), one may alter the asymptotic ansatz for the volume force $f(h, x)$. Indeed, if the longitudinal components $f_{z_\pm}(h, x) = O(h^{-2})$ happen to be much bigger than the transversal components $f_{y_\pm}(h, x) = O(h^0)$, then the above-mentioned differential equations for $w_1^{0,\pm}$ gain non-trivial right-hand sides so that the averaged longitudinal displacements are no longer constant. The same formula $w_1^{0,\pm} \neq \text{const}$ occurs by supposing that the traction force $g(h, x)$ at the surface $\gamma_\pm(h)$ of the knots $\mathcal{G}_h^\pm$ also get different orders in h for longitudinal and transversal components.

The assumptions on the external forces described above become very realistic in one particular example of the junction $\Omega(h)$ depicted in Figure 1 and considered in [5,6]. Let one leg, say $Q^+(h)$, be clamped at its exterior knot $\mathcal{G}_h^+$ and let the volume force be absent. Furthermore, the traction force applied at $\gamma_-(h)$ has only the longitudinal component. In this case the ansatz (4.2) gives the main asymptotic terms of the elastic fields in the whole junction, while the ansatz (2.1)–(2.3) works for the clamped leg only because of the “far-interaction” effect mentioned in the end of Section 1.1.

4.2 | An example of shape optimization

We now consider the optimization of the junction $\Omega(h)$ with respect to the angle $\alpha \in (0, \pi)$. We assume that $f = 0$, $g^0 = 0$ and the principle vectors

$$\mathbf{G}^\pm = e_1 \mathbf{G}_1^{\pm} + e_1 \mathbf{G}_2^{\pm}, \quad |\mathbf{G}^+| = |\mathbf{G}^-| =: \mathbf{G}$$

of the traction forces $g^\pm$ act in opposite directions along the line $\mathcal{L}$ connecting the points P^+ and P^- , see Figure 3(a).

We select the angle $\alpha = \alpha_*$ such that the deflection of the beams becomes maximal. We denote by $\varphi_\pm(\alpha)$ the acute angles between the line $\mathcal{L}$ and the axes $z_\pm$ (see Figure 3(a))

$$\sin \varphi_\pm(\alpha) = \sin \alpha \, l_\mp (l_+^2 + l_-^2 - 2l_+ l_- \cos \alpha)^{-1/2}. \quad (4.6)$$

According to (2.41) and (2.40), the main part of the deflection is defined by the formula

$$|w_2^\pm(z_\pm)| = |\mathbf{G} \sin(\varphi_\pm(\alpha)) (l_\pm z_\pm^2/2 - z_\pm^3/6), \quad z_\pm \in (0, l_\pm). \quad (4.7)$$

Let $l_+ > l_-$. In view of (4.6), it is easy to verify that $\sin \varphi_\pm(\alpha)$ attend maximum value when $\alpha = \alpha_*$ with $\alpha_* = \arccos(l_-/l_+)$. By (4.6), we have

$$\sin \varphi_-(\alpha_*) = 1, \quad \sin \varphi_+(\alpha_*) = l_-/l_+. \quad (4.8)$$

The function $|w_2^\pm|$ attains maximum at the ends $z_\pm = l_\pm$ of the beams. Due to (4.7) and (4.8) we conclude that

$$\max_{\alpha \in (0, \pi)} |w_2^\pm(z_\pm)| = l_+^2 l_- / 3.$$

The maximum is attained at the end $z_+ = l_+$ of the longest beam $\mathcal{Q}_+(h)$.

4.3 | Junctions with several beams

In order to clarify the difference between the ansätze (1.1) and (1.2), we consider two elastic structures, namely Ω_1 with three beams $\mathbf{G}^I, \mathbf{G}^{II}, \mathbf{G}^{III}$ in Figure 3(b) and Ω_2 with four beams $\mathbf{G}^I, \dots, \mathbf{G}^{IV}$ in Figure 3(c). The skeleton $\mathbf{S}_1$ ($\mathbf{S}_2$) of the junction Ω_1 (Ω_2) consists of four points $\mathbf{P}_k$ and three (four) segments l^K .

We suppose that all beams are made from the same isotropic elastic material with the Lamé constants λ and $\mu > 0$. The constant traction force $\mathbf{g}$ applied to the beam $\mathbf{G}^I$ in both cases. All other beams are not loaded. Let $z^K \in (-1, 1)$, $K = I, II, III, IV$ be the local longitudinal coordinate on the segment l^K , $z^I = -1$ for $\mathbf{P}_0$, $z^I = 1$, $z^{II} = -1$ for $\mathbf{P}_1$, $z^{II} = 1$, $z^{III} = -1$ for $\mathbf{P}_2$, $z^{III} = 1$ for $\mathbf{P}_3$ and $z^{IV} = -1$ for $\mathbf{P}_0$, $z^{IV} = 1$ for $\mathbf{P}_2$.

For the junction Ω_1 , we use ansatz (1.2) on $\mathbf{G}^I, \mathbf{G}^{III}$ and (1.1) on $\mathbf{G}^{II}$ (cf. [14,16]). The functions w_2^K and constant a^{II} satisfy ordinary differential Equations 4.3 with $\mathcal{M}_{22}^\pm = \mu(\lambda + \mu)/(3(\lambda + 2\mu)) =: \mathbf{m}$, $\mathbf{F}_2^\pm = \mathbf{m}^{-1}\mathbf{g}/24$ and thirteen conditions at the points $\mathbf{P}^K$. The boundary conditions at the external ends, cf. (1.3), and transmission conditions, cf. (1.4), (1.5), read:

$$\begin{aligned} w_2^I(-1) &= \partial_z w_2^I(-1) = w_2^{III}(1) = \partial_z w_2^{III}(1) = 0, w_2^{II}(-1) = w_2^{II}(1) = 0, \\ \partial_z w_2^I(1) &= \partial_z w_2^{II}(-1), \partial_z w_2^{II}(1) = \partial_z w_2^{III}(-1), \\ \mathbf{m}(\partial_z^2 w_2^I(1) - \partial_z^2 w_2^{II}(-1)) &= 0, \mathbf{m}(\partial_z^2 w_2^{II}(1) - \partial_z^2 w_2^{III}(-1)) = 0, \\ -w_2^I(1) &= w_2^{III}(-1) = a, \mathbf{m}(\partial_z^3 w_2^I(1) + \partial_z^3 w_2^{III}(-1)) = 0. \end{aligned} \quad (4.9)$$

We impose homogeneous boundary and transmission conditions

$$\begin{aligned} w_1^I(-1) &= \partial_z w_1^I(-1) = w_1^{III}(1) = \partial_z w_1^{III}(1) = 0, \\ \partial_z w_1^I(1) &= \partial_z w_1^{II}(-1), \partial_z w_1^{II}(1) = \partial_z w_1^{III}(-1). \end{aligned} \quad (4.10)$$

The system (2.18), (4.9)–(4.10) with the conditions (1.11) for w_1^{II} has the solution $w_1^K = 0$, $K = I, II, III$,

$$\begin{aligned} w_2^I &= \mathbf{g}(-21z^4 + 49z^3 + 37z^2 - 157z^1 - 124)/21, w_2^{II} = \mathbf{g}(-24z^3 - 14z^2 + 24z^1 + 14)/21, \\ w_2^{III} &= \mathbf{g}(35z^3 + 19z^2 - 143z^1 + 89)/21, a = 72\mathbf{g}/7. \end{aligned} \quad (4.11)$$

All beams in the junction Ω_2 are hard-movable according to [14] and we use ansatz (1.2) on $\mathbf{G}^K$, $K = 1, 2, 3, 4$. For the junction Ω_1 , we use the boundary conditions (4.9)_{1,2} and the transmission conditions

$$\begin{aligned} w_2^I(1) &= w_2^{III}(1) = w_2^{IV}(1) = 0, \partial_z w_2^{II}(1) = \partial_z w_2^{III}(-1) = \partial_z w_2^{IV}(1), \\ w_2^{IV}(-1) &= \partial_z w_2^{IV}(-1) = 0, \mathbf{m}(\partial_z^2 w_2^{II}(1) - \partial_z^2 w_2^{III}(-1) + \partial_z^2 w_2^{IV}(1)) = 0. \end{aligned} \quad (4.12)$$

The solution of the obtained problem is significantly different from the solution of system (2.18), (4.9), (4.10). The main asymptotic term of the displacement of the upper-right point $\mathbf{P}_2$ equals to $h^{-2}6\mathbf{g}$ for junction Ω_1 (see (4.11)). However, according to (4.12) in the case of junction Ω_2 , the displacement of this point does not admit the terms of the order $O(h^{-2})$. The reinforcement of the construction by the additional beam $\mathbf{G}^{IV}$ is described by the modified ansatz (1.1).

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